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Ediacaran paleogeography of Laurentia: Paleomagnetism and ^{40}Ar–^{39}Ar geochronology of the 583 Ma Baie des Moutons syenite, Quebec

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ABSTRACT

Laurentia has unclear paleogeographic relations during the Precambrian–Cambrian transition. Published paleomagnetic results from the late Neoproterozoic (Ediacaran period) imply that Laurentia may have lain at high southerly latitudes during 590–570 Ma, but conflict with other results from 615 Ma and from 565 to 550 Ma which place Laurentia at low paleolatitudes. Paleomagnetic results from other continents over the same Ediacaran period are either sparse or equally ambiguous, giving rise to proposals of unusual geodynamic events such as rapid plate motions and/or large-scale true polar wander or unusual geomagnetic field geometry. To address this problem for Laurentia, we have obtained paleomagnetic and ^{40}Ar–^{39}Ar geochronologic results from the mid-Ediacaran Baie des Moutons (Mutton Bay) syenite, exposed in a failed Iapetan rift along the north shore of the Gulf of St. Lawrence in eastern Quebec. Early and late intrusive units of the Mutton Bay syenite bear hornblende and biotite grains which provide overlapping ^{40}Ar–^{39}Ar plateau ages, indicating that the intrusion cooled rapidly upon emplacement at 583.4 ± 2.0 Ma. Of the 44 sites sampled in the Baie des Moutons syenite, 8 yielded consistent characteristic paleomagnetic ChRM A directions that were easterly and steep ($D = 98.6^\circ$ $I = 78.0^\circ$; $\alpha_{95} = 6.5^\circ$, $k = 71.7$), retained by PSD magnetite. Six sites from associated feldspar porphyry dykes and late intrusive units gave shallow southeasterly or northwesterly ChRM B directions ($D = 163.1^\circ$, $I = 6.0^\circ$; $\alpha_{95} = 21.7^\circ$, $k = 10.5$). Other sites fail to carry a detectable or stable ancient remanence, mainly due to the dominant presence of a viscous remanence carried by MD magnetite. Baked contact tests for both ChRMs A and B at several sites proved inconclusive due to the lack of stable remanence in the host syenite in each case. The eight sites bearing ChRM A directions yield a paleopole at 42.6°N, 332.7E ($dp = 11.7^\circ$, $dm = 12.4^\circ$) which is close to three 590–570 Ma previously published paleopoles that also place Laurentia at high paleolatitudes. The six sites with ChRM B directions give a paleopole at location −34.2°N, 321.5E ($dp = 10.9^\circ$, $dm = 21.8^\circ$), in good agreement with a published paleopole from the 565 Ma Sept Iles intrusion. The results from this study are enigmatic, providing support for both high and low paleolatitude interpretations of Laurentia's location during the mid-Ediacaran. It is nevertheless possible to propose relative Ediacaran paleogeographies that are consistent with Ediacaran paleopoles and with later Cambro-Ordovician relationships between Laurentia and other circum-Iapetus continents. The candidate paleogeographies do require, however, some contribution from rapid plate motion or true polar wander and also indicate that the opening of the Iapetus Ocean began well before 550 Ma rifting along eastern Laurentia.

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1. Introduction

Global paleogeography is poorly known prior to Cambrian time. For most of the Paleozoic, numerous paleomagnetic results coupled with the existence of supercontinents such as Gondwana and then Pangea guide the reconstruction of paleogeography (Van der Voo, 1993; Stampfli and Borel, 2002). Index fossils have additionally proven useful for constraining biostratigraphic age and for indicating the faunal provinciality of rocks. Such paleogeographic tools are not, however, as readily available for the Precambrian (Waggoner, 1999), which additionally tends to suffer from having far fewer well-determined, well-dated paleomagnetic results (Meert and Torsvik, 2003).

The late Precambrian to Cambrian appears to have been a time of profound transition for the Earth, being host to the burgeoning of metazoan life (Grotzinger et al., 1995; Knoll et al., 2004; Budd, 2008) and to severe extremes in climate (Hoffman and Schrag, 2002). During Neoproterozoic time, widespread rifting and

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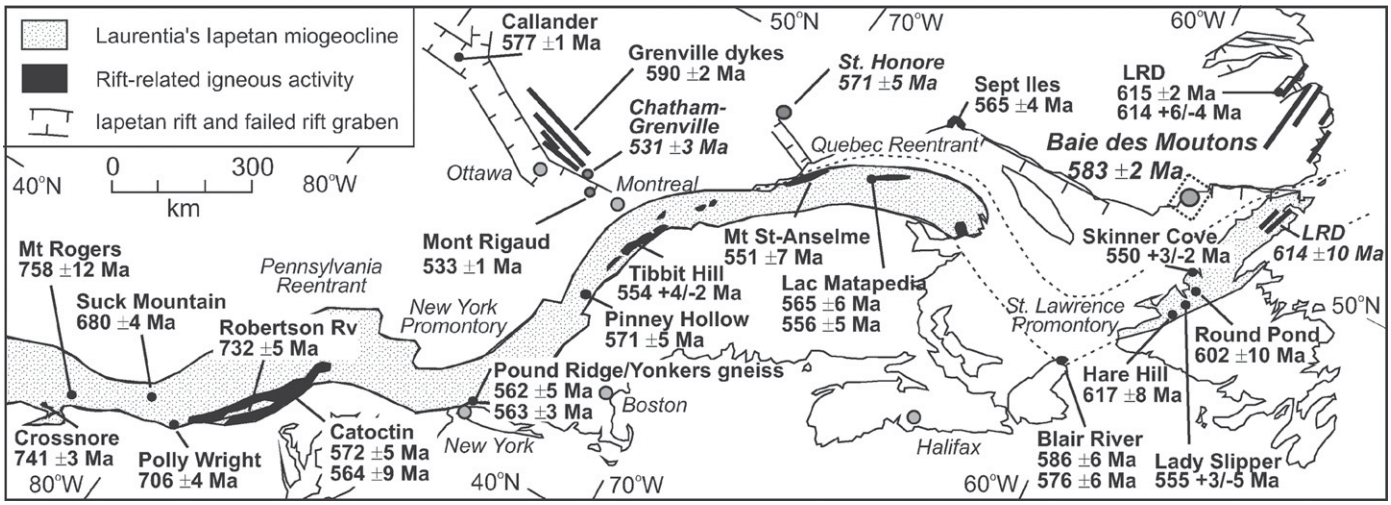


Fig. 1. Distribution of Neoproterozoic to Cambrian rift-related magmatic rocks along the incipient Iapetan margin of Laurentia, in present-day eastern North America (after Cawood et al., 2001). The Baie des Moutons syenite in eastern Quebec is highlighted and enclosed by a box (dotted line) indicating the area of Fig. 2a. Ages given in italics are determined by the ^{40}Ar–^{39}Ar method; all others are by U–Pb. Age errors age quoted as 2σ. LRD refers to the Long Range dykes of southeastern Labrador and northern Newfoundland. Unit ages are drawn from the following references: *Baie des Moutons complex* – this study; *Blair River mategabbro dykes* – Miller and Barr (2004); *Callander complex* – Kamo et al. (1995); *Catoctin Formation metavolcanics and dykes* – Aleinikoff et al. (1995); *Chatham-Grenville stock* – McCausland et al. (2007); *Crossnore pluton* – Su et al. (1994); *Grenville dykes* – Kamo et al. (1995); *Hare Hill granite* – van Berkel and Currie (1988); *Lac Matapedia volcanics* – Cox and Hodych (2005); *Lady Slipper pluton* – Cawood et al. (1996); *Long Range dykes* – Kamo et al. (1989), Kamo and Gower (1994), Stukas and Reynolds (1974); *Mont Rigaud stock* – McCausland et al. (2007); *Mount Rogers rhyolite* – Aleinikoff et al. (1995); *Mt. St-Anselme volcanics* – Cox and Hodych (2005); *Pinney Hollow Formation metafelsite* – Walsh and Aleinikoff (1999); *Polly Wright Cove pluton* – Tollo et al. (2004); *Robertson River batholith* – Lukert and Banks (1984); *Round Pond granite* – Williams et al. (1985); *Sept Iles layered intrusion* – Higgins and van Breemen (1998); *Suck Mountain pluton* – Tollo et al. (2004); *Tibbit Hill Formation metavolcanics* – Kumarapeli et al. (1989); *Pound Ridge granite gneiss and Yonkers gneiss* – Tollo et al. (2004).

orogenic activity accompanied the breakup of a supercontinent, Rodinia, the opening of new ocean basins such as the Iapetus, and the assembly of the supercontinent Gondwana (Hoffman, 1991; Meert and Van der Voo, 1996; Dalziel, 1997; Tohver et al., 2006; Li, 2008).

In this evolving paleogeography, Laurentia is envisaged to have occupied a 'keystone' position at the core of the ca. 1 Ga Rodinia supercontinent, and to have become more isolated from other continents as rifting along its margins occurred through the mid to late Neoproterozoic (Hoffman, 1991; Cawood et al., 2001; Li, 2008). Paleomagnetic data for Laurentia are sparse to non-existent during much of the late Neoproterozoic, however. Existing paleomagnetic data imply rapid latitudinal motions for Laurentia from low latitudes to the south pole between 615 and 590 Ma and then from the south pole to near the equator between 570 and 550 Ma (McCausland et al., 2007). Such rapid apparent motions complicate attempts to reconstruct the paleogeography, and have led to non-uniformitarian proposals such as rapid plate motion during the terminal Neoproterozoic opening of the Iapetus (McCausland and Hodych, 1998; Cawood et al., 2001), contributions from large-scale true polar wander (Evans, 1998, 2003) or equatorial dipole geomagnetic field geometry (Abrajevitch and Van der Voo, 2010).

In this work, we provide paleomagnetic results and new hornblende, biotite and potassium feldspar ^{40}Ar–^{39}Ar ages for three intrusive phases of the mid-Ediacaran Baie des Moutons (Mutton Bay) syenite complex of the Quebec north shore, Gulf of St. Lawrence (Lalonde and Martin, 1983). Our investigation of the Baie des Moutons complex is motivated by the need to test the mid-Ediacaran position of Laurentia with a new, well-dated paleomagnetic result: did Laurentia reside at high paleolatitudes during mid-Ediacaran time? The complex is emplaced in cratonic Laurentia and so the paleomagnetic data apply directly to Laurentia. Our ^{40}Ar–^{39}Ar mineral ages define the cooling history of the complex, both to refine its mid-Ediacaran age and to help pin down the age of remanence acquisition.

2. Geological setting, sampling and previous work

The rift-related magmatism in eastern Laurentia (Fig. 1) is present in two age pulses, a volumetrically minor ca. 750 Ma largely felsic group in the south and central Appalachians and a major latest Neoproterozoic pulse in the central to northeastern Appalachians (Aleinikoff et al., 1995; Cawood et al., 2001). The younger pulse consisted of dominantly tholeiitic magmatism between 615 and 590 Ma that gave way to rift-setting alkalic to carbonatitic intrusives, heralding the development of Laurentia's early Paleozoic Iapetan passive margin (Kumarapeli and Saull, 1966; Kumarapeli, 1993; Cawood et al., 2001).

The 20 km diameter Baie des Moutons syenite complex is prominently emplaced in Grenville gneiss along the Quebec north shore of the Gulf of St. Lawrence (Davies, 1968; Gerencher, 1967; Lalonde, 1981; Lalonde and Martin, 1983), as part of the margin-wide rifting activity during the late Neoproterozoic (Figs. 1 and 2). Detailed mapping by Davies (1968) of the Baie des Moutons complex and environs revealed 23 identifiable intrusive units ranging from syenitic to quartz–syenitic composition, along with numerous syn-intrusive microsyenite and aplite dykes and 'late' intrusive mafic or carbonatitic dykes. The overall ovoid shape of the intrusion, its distribution of radial and concentric dykes and the general centreward dip of intrusive contacts between units indicates the complex to have the form of a set of nested cones, typical of anorogenic, rift-setting intrusions. The distribution of dyke orientations, having steep dips and varied strikes (Gerencher, 1967) suggests that the host intrusive complex has not been significantly tilted since its emplacement. Gerencher (1967) did a preliminary paleomagnetic study of specimens from dykes from the Baie des Moutons complex, finding that some of the units bore stable remanence up to 40 mT demagnetization.

Lalonde (1981) and Lalonde and Martin (1983) grouped the intrusive units recognized by Davies (1968) by contact relations

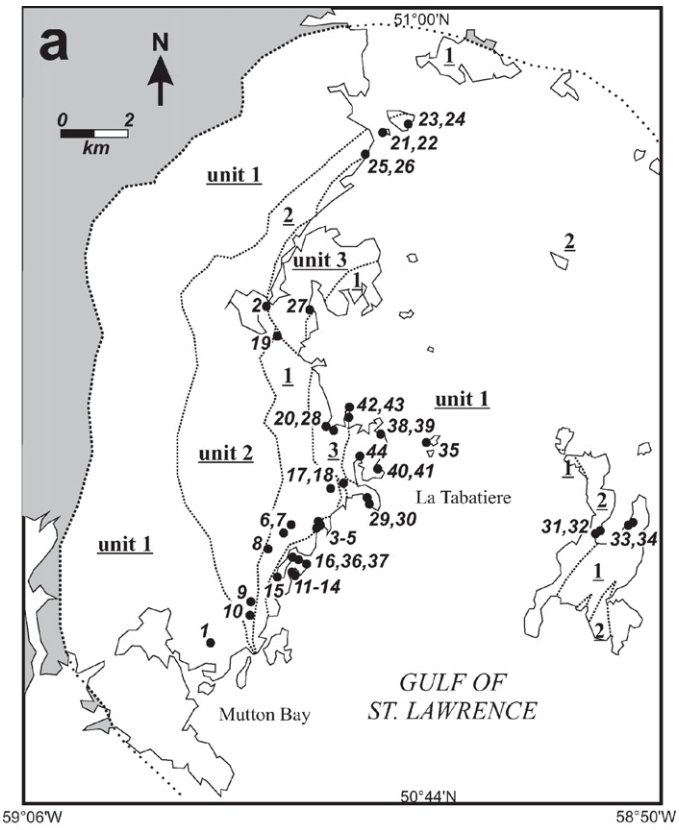


Fig. 2. a) Geology and sampling sites for the Baie des Moutons syenite complex, modified from Lalonde and Martin (1983) and Lalonde (1981). Intrusive units 1, 2 and 3 are indicated in underlined text. ^{40}Ar–^{39}Ar mineral ages obtained in this study are shown for three sites; b) glacially polished shoreline outcrop of Baie des Moutons site MB16, a red feldspar porphyry dyke, seen as a recessively weathered 2 m wide slot in the massive syenitic host.

and composition into three main intrusive units (Fig. 2a). They contain a progression of intrusive units from dark green feldspar porphyry-hornblende syenite to grey hornblende syenite to trachytic quartz syenite, reflecting broadly increasing silica content during the evolution of a parental magma (Lalonde and Martin, 1983). All units are cut by aplite and microsyenitic dykes that have no discernable intrusive order from their apparently mutual cross-cutting relations. The three main intrusive units and their aplitic and syenitic dykes are cut by co-magmatic red north-northeast trending feldspar porphyry dykes (Fig. 2b) and minor carbonatite dykes that are the last magmatic event for the Baie des Moutons complex (Lalonde and Martin, 1983).

A whole rock K–Ar age of 575 Ma was determined for the Baie des Moutons complex by Doig and Barton (1968; recalculated using the decay constant of Steiger and Jaeger, 1977), indicating the intrusion to likely be a part of the late Neoproterozoic rift-related magmatism which affected northeastern North America (Doig and Barton, 1968; Kumarapeli and Saull, 1966).

Forty-four paleomagnetic sites were sampled from excellent exposures of the complex and associated NNE dykes (Fig. 2), spanning the duration of intrusive events in the Baie des Moutons syenite. Three of these sites were dated using hornblende, biotite and potassium feldspar ^{40}Ar–^{39}Ar geochronology to bracket the timing of magmatism and remanence acquisition for units in the complex.

Sites were mostly sampled in glacially exposed outcrops accessible by small boat from La Tabatiére (Fig. 2b); others were taken from roadcut outcrops onshore. Paleomagnetic baked contact tests (Graham, 1949) were sampled where intrusive units of the complex showed definitive contact relations. Each site typically consisted of six oriented 2.54 cm diameter core samples, obtained by portable gas-powered drill. Cores were sliced into 2.2–2.4 cm length specimens. In all, the 44 Baie des Moutons sites provided 530 specimens for analysis.

3. Paleomagnetism

3.1. Definition of magnetic mineralogy and characteristic remanent magnetization (ChRM)

Specimens from the Baie des Moutons syenite complex all underwent magnetic susceptibility measurement, and selections from these underwent low temperature liquid N_2, thermal and/or alternating field (AF) stepwise demagnetization at the University of Michigan. Magnetic hysteresis and backfield parameters (M_{rs}/M_s; H_{cr}/H_c) were measured at the Institute for Rock Magnetism, University of Minnesota. Initial magnetic susceptibility measurements were made using a dual frequency Bartington magnetic susceptibility meter and a KLY-1 Kappabridge. Further measurements of magnetic susceptibility variation with temperature for selected samples were made at the University of Windsor with a KLY-3 Kappabridge equipped with a furnace. Demagnetizations and remanence measurements were made at the University of Michigan using a Sapphire Instruments SI-4 AF demagnetizer, an ASC TD-48 thermal demagnetizer, and a 2G cryogenic magnetometer, all set inside a magnetically shielded room which permits entry of only ~0.2% of the Earth's ambient magnetic field intensity.

Magnetic susceptibility ranged over more than two orders of magnitude for Baie des Moutons sites (Table 1), from 20×10^{-6} SI units for aplite and microsyenite dykes and fine grained red syenite units to over 5000×10^{-6} SI for some coarse syenite units, appearing to reflect the presence of high susceptibility multidomain magnetite. In a modified Day plot (Dunlop, 2002), samples from Baie des Moutons sites plot in the pseudo-single domain (PSD) magnetite and multidomain (MD) magnetite fields, indicating that the syenites can be interpreted to contain PSD–MD magnetite mixtures, with little or no contribution from other magnetic minerals.

Magnetic suceptibility with heating and cooling in air (Fig. 3) of Baie des Moutons syenite and dyke samples similarly indicate a characteristic loss of susceptibility between 550 and 580 °C, reflecting the Curie temperature $T_C = 580$ °C of Ti-poor magnetite. Two syenites and an aplite dyke from early intrusive phases of Baie des Moutons show relatively simple, reversible heating and cooling curves, indicating that no new magnetic minerals were created during heating in air (Fig. 3a–c). Two samples from late-stage dykes in the Baie des Moutons intrusion, however, show a susceptibility increase reflecting the development of new magnetic minerals during heating. A north-northeast red feldspar porphyry dyke sample

Table 1
Site mean remanence data for the 583 Ma Baie des Moutons syenite, Quebec (50°50.4′N, 58°57.6′W).

Site	Rocktype	Intr. unit	χ	NRM mA/m	N	$e+g$	ChRM direction					Site location	
			10^{-6} SI				Dec (°)	Inc (°)	α_{95} (°)	k	Type	50°N+	58°W+
MB01[a]	Fsp-por dyke	d	2819	1000	21	–						46.68′	59.99′
MB02	Syenite xeno	3	1250	151	9	–						52.13′	59.91′
MB03	Fol grey syenite	2	1862	196	13	10+1	109.2	80.8	7.2	41.5	A	48.44′	58.62′
MB04	Fol grey syenite	2	1951	211	13	7+2	82.9	79.4	7.1	54.7	A	48.65′	58.61′
MB05	Pink fsp syenite	1	1917	3240	8	5+0	141.5	82.8	9.3	68.8	A	48.68′	58.63′
MB06	Aplite dyke	1	31	3	9	7+2	183.8	−5.9	5.9	80.1	B	48.42′	59.49′
MB07	Fsp-por dyke	d	829	80	10	–						48.48′	59.45′
MB08[a]	Green syenite	1	1577	251	14	–						48.37′	59.88′
MB09	Grey syenite	2	2177	169	9	–						47.39′	59.97′
MB10	Aplite dyke	2	1661	335	12	–						47.21′	59.98′
MB11	Red fsp dyke	d	43	5	9	4+1	162.8	−3.7	4.4	325	B	47.83′	59.35′
MB12	Red fsp dyke	d	20	4	9	3+2	163.2	23.7	9.1	92.1	B	47.83′	59.35′
MB13	Syenite dyke[c]	2	3634	413	14	–						47.83′	59.35′
MB14	Syenite dyke[c]	2	2806	265	18	–						47.83′	59.35′
MB15	Grey syenite[c]	2	193	19	19	0+5	94.5	64.4	–	10.8	A	47.75′	59.62′
MB16[a]	Red fsp dyke[c]	d	102	2	11	–						48.09′	59.35′
MB17	Aplite dyke	1	5515	1630	10	–						49.29′	58.30′
MB18[a]	Red syenite	3	218	417	10	–						49.38′	58.05′
MB19	Aplite dyke	1	2881	404	8	5+0	46.4	73.6	11.4	46.1	A	51.64′	59.67′
MB20	Red syenite	3	703	121	10	9+0	321.1	22.0	6.7	59.6	−B	50.20′	58.56′
MB21	Aplite dyke	2	5495	403	10	–						54.96′	57.01′
MB22	Aplite dyke[c]	2	2652	1430	7	–						54.96′	57.01′
MB23	Finer granite	2	4224	562	9	–						55.14′	56.34′
MB24	Finer granite	2	3600	504	9	–						55.14′	56.34′
MB25	Aplite + host	2	514	91	8	5+0	78.7	79.7	9.7	62.9	A	54.61′	57.53′
MB26	Mafic dyke[c]	3	3360	258	5	5+0	136.0	80.7	8.5	81.5	A	54.61′	57.53′
MB27[b]	Green syenite	3	625	294	8	1+0	80.1	57.4	–	–		52.10′	58.88′
MB28	Xeno red syenite	3	3343	390	12	–						50.12′	58.44′
MB29	Fol red syenite	3	1388	303	12	–						48.97′	57.45′
MB30	Xeno in syenite	3	1940	519	15	–						48.99′	57.50′
MB31	Fol grey syenite	2	2918	1130	12	–						48.65′	51.78′
MB32[a]	Fol grey syenite	2	2215	396	12	–						48.62′	51.88′
MB33	Pink fs syenite	1	2267	291	12	–						48.80′	51.05′
MB34	Aplite dyke	1	97	192	17	–						48.79′	51.06′
MB35	Aplite dyke	1	43	58	16	–						50.05′	56.01′
MB36	Red fsp dyke[c]	d	77	4	10	9+0	173.1	9.7	10.3	25.8	B	48.09′	59.17′
MB37	Red fsp dyke	d	209	18	13	–						48.00′	58.97′
MB38	Fol green syenite	1	160	28	14	–						50.00′	57.02′
MB39	Fol green syenite	1	26	2	12	–						49.98′	57.00′
MB40	Green syenite	1	44	3	18	–						49.52′	57.30′
MB41	Green syenite	1	208	24	13	–						49.53′	57.29′
MB42[b]	Fol green syenite	1	894	75	13	–						50.45′	57.96′
MB43	Carbonatite	d	201	36	9	7+0	151.4	33.4	5.8	108	B	50.48′	57.96′
MB44	Green syenite	1	193	23	13	7+3	124.0	69.2	7.3	46.0	A	49.69′	57.75′
ChRM A 8 'steep' sites				**8**		**98.6**		**78.0**		**6.6**	**71.7**		A
ChRM B 4 'shallow' dyke sites				4		170.8		6.0		16.9	23.3		B4
ChRM B 5 'shallow' dyke sites including site 43				5		167.4		11.6		15.6	20.0		B5
ChRM B all 6 'shallow' sites, including site 20				**6**		**163.1**		**6.0**		**21.7**	**10.5**		B

Rocktype: fol = foliated, fsp = feldspar, por = porphyry, xeno = syenite xenolith; intr. unit: identity of site within the Baie des Moutons intrusive unit order of 1–3 or d (dykes); NRM: mean of specimen natural remanent magnetization intensities in the site, in units of mA/m; N: total number of specimens demagnetized in site; $e+g$: number of endpoint (e) and great-circle (g) specimens useable for site mean ChRM calculation. Site mean ChRM directions are given by: Dec – declination; Inc – inclination; α_{95} – radius of cone of 95% confidence, and k – precision parameter estimate. ChRM type for each site with enough contributing specimens is given as "A" or "B," as discussed in text. Site location in NAD-27 coordinates is given as the decimal minutes in addition to 50°N and 58°W.

[a] Site contains two or more ChRM-bearing specimens that have inconsistent directions.
[b] Site has too few ChRM-bearing specimens to calculate a site mean direction.
[c] Site of an attempted, but inconclusive baked contact test.

from site MB36 (Fig. 3d) is dominantly paramagnetic with a slight drop at 580 °C, followed upon cooling by a marked increase in magnetic susceptibility at 580 °C. The carbonatite dyke sample from site MB43 (Fig. 3e) shows an increase in magnetic susceptibility at 580 °C upon cooling as well, but also an increase in susceptibility during heating above 400 °C which likely corresponds with the growth of a new magnetic mineral that appears to have a separate, somewhat lower Curie temperature of T_C = 530 °C. The dyke samples with such non-reversable behaviour suggest that for these sites, the interpretation of thermal demagnetization data at high temperature steps should be considered with caution above ~400 °C.

Measurement of the specimens' natural remanent magnetizations (NRM) showed intensities which spanned more than three orders of magnitude, from ~2 to 3240 mA/m (Table 1). Unit 1 dark green syenite specimens and those specimens from aplite dykes and the red feldspar porphyry late dykes tended to have weak NRM intensities (1–100 mA/m; Table 1) whereas specimens from coarse syenite units and their xenoliths typically had stronger NRM intensities.

After NRM measurement, representative pilot specimens were either AF demagnetized in 12 steps up to 120 mT or thermally demagnetized in 14 steps up to 580 °C. Most specimens lost between 50% and 95% of their NRM intensity by 20 mT or 450 °C,

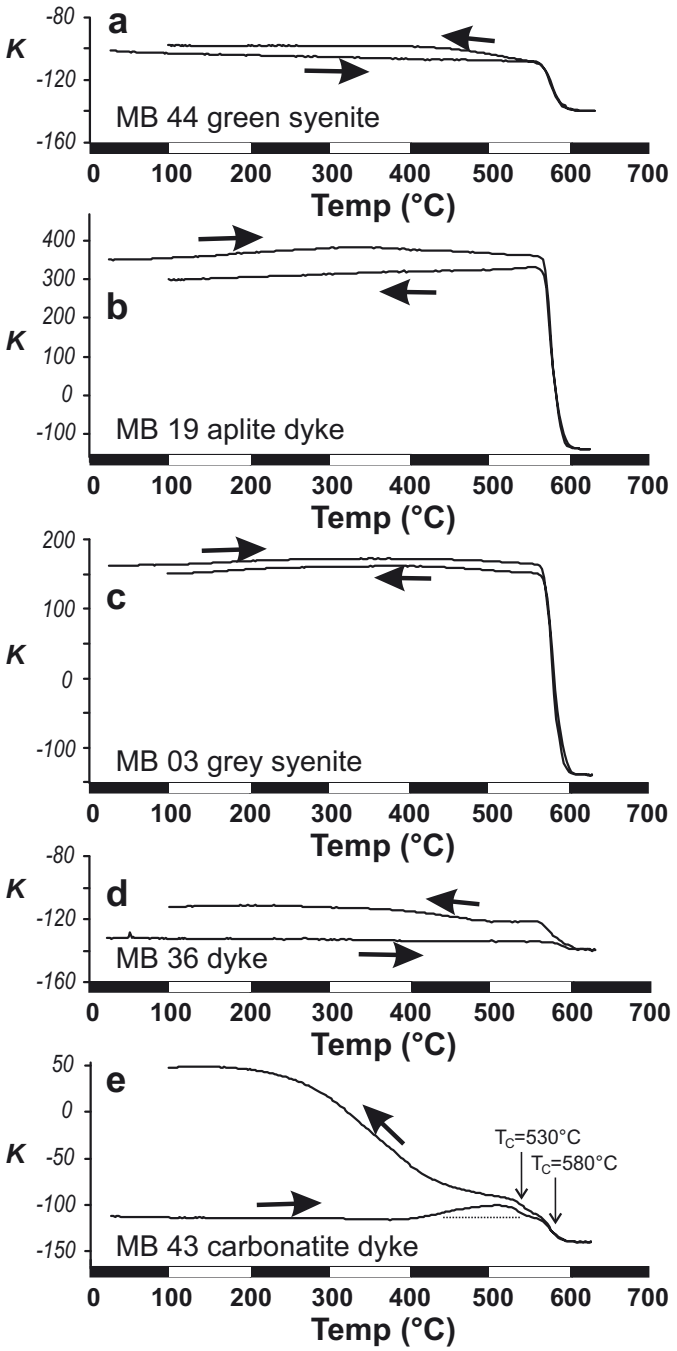


Fig. 3. Magnetic suceptibility curves in 10^{-6} SI units for the heating and cooling (as indicated by arrows) of Baie des Moutons syenite and dyke samples over temperature ranges similar to those encountered during standard thermal demagnetization. All samples showed a characteristic loss of susceptibility between 550 and 580 °C, indicating low-titanium magnetite to be the dominant magnetic carrier present. Samples a–c are of syenites and an aplite dyke representing early intrusive phases of Baie des Moutons; they show relatively simple, reversible heating and cooling curves, indicating that no new magnetic minerals were created during heating. Samples d and e are from late-stage dykes in Baie des Moutons; both show a susceptibility increase reflecting the development of new magnetic minerals during heating. The site MB43 carbonatite dyke sample in particular exhibits an increase in susceptibility with heating above 400 °C (marked by the departure from the expected flat dotted line) which likely corresponds with the growth of a new magnetic mineral that appears to have a separate, somewhat lower Curie temperature of T_C = 530 °C.

indicating MD magnetite to be a significant remanence carrier in most sites. Multidomain magnetite is of particular concern because it can acquire remanence at temperatures as low as ~200 °C, but may require laboratory unblocking temperatures of ~450 °C to remove the same remanence (Dunlop and Argyle, 1991) thus significantly overlapping with the useful remanence information carried by pseudosingle and single domain magnetite.

Following these observations of the pilot specimens' demagnetization behaviour, other specimens were "AF-cleaned" in three to five demagnetization steps up to 20 mT, and then stepwise thermally demagnetized to reveal their characteristic remanent magnetizations (Fig. 4). Some specimens underwent cleaning by low temperature demagnetization (LTD; Dunlop and Argyle, 1991; Dunlop and Özdemir, 1997) via cycling the specimens through the Verwey transition of magnetite in a field-free space using a liquid N_2 bath, requiring at least three liquid N_2 cycles to remove most of the viscous remanence. Specimens undergoing LTD cycling were placed in different orientations during each 1 h liquid N_2 bath cycle, and then allowed to warm to room temperature prior to measurement. Each LTD cycle took place within a pyrex glass dish, inside a styrofoam insulated container within a mu-metal shield in the magnetically shielded room at the University of Michigan.

Characteristic remanent magnetization (ChRM) directions were determined using specimens that exhibited stable remanence at unblocking temperatures of >450 °C and coercivities of >20 mT in cases where the specimens exhibited a substantial VRM "soft" magnetization component. For low-NRM intensity specimens with only minor or no VRM contribution, ChRM could be defined at >15 mT and at unblocking temperatures of >260 °C. ChRM directions were identified using the least-squares line-fitting method over four or more consecutive AF or thermal steps (Kirschvink, 1980) where unbiased demagnetization trends could be clearly defined with maximum angular deviation of <15°, and by great-circle analysis (McFadden and McElhinny, 1988) where some portion of the viscous remanence overlapped the ChRM.

3.2. Paleomagnetic results

Fig. 4 sets out examples of characteristic demagnetization behaviour for specimens from the Baie des Moutons syenite complex. A northerly and downward-directed viscous remanence was apparent in many specimens and was typically removed by AF demagnetization steps to 20 mT or by three to four liquid N_2 baths (Fig. 4). This viscous remanent magnetization (VRM) has an overall mean direction of D = 1.4°, I = 75.1°; α_{95} = 1.2°, k = 47 (n = 293 specimens) which is similar to the expected geocentric axial dipole (GAD) field direction of 0°, 69°, and to the IRGF 11 model present-day earth field (PEF) direction of 336.4°, 72.7° for the Baie des Moutons locality during sampling (Fig. 5a). The VRM in Baie des Moutons specimens is carried by multidomain magnetite and is most likely of recent origin. In several sites, particularly those with greater NRM intensity, the VRM was the dominant component, contributing more than 90% of the NRM (Fig. 4a and l), rendering the identification of possible ChRMs difficult.

ChRM directions were isolated in specimens from Baie des Moutons sites using AF demagnetization to 15 mT or 20 mT (AF "cleaning") or LTD, followed by thermal demagnetization to at least 450°. Two ChRM directional groups were found in the Baie des Moutons syenite complex (Table 1; Fig. 5b). A steep down-directed ChRM "A" occurs in specimens from eight syenite sites, defined over unblocking temperatures >450 °C and at coercivities >20 mT. ChRM A tends to decay progressively towards the origin of orthogonal vector plots (Fig. 4b, c, d), often showing a sharp intensity drop in the 550–575 °C range, consistent with PSD magnetite as the remanence carrier. In six sites sampled from "late" co-magmatic dykes and syenitic units a ChRM "B" was found, directed shallowly either

to the south-southeast or to the northwest (Fig. 4e, f, g, h). ChRM B occurs in specimens from sites with low magnetic susceptibility and NRM intensity (Table 1). In the NNE-trending red feldspar porphyry dykes (sites 11, 12, 36) ChRM B tends to be removed over unblocking temperatures of between 260° and 450–520 °C (Fig. 4f), somewhat lower than the unblocking range of ChRM A. The two ChRMs are nowhere observed to coexist.

3.3. Low temperature demagnetization (LTD) effect upon ChRM

In this study, low temperature demagnetization (LTD; Dunlop and Argyle, 1991; Dunlop and Özdemir, 1997) pre-treatment was found to remove much of the VRM from Baie des Moutons specimens, in a manner that showed greater directional stability than that found in AF "cleaning" to 20 mT (Fig. 4i, j, k, l). We also found,

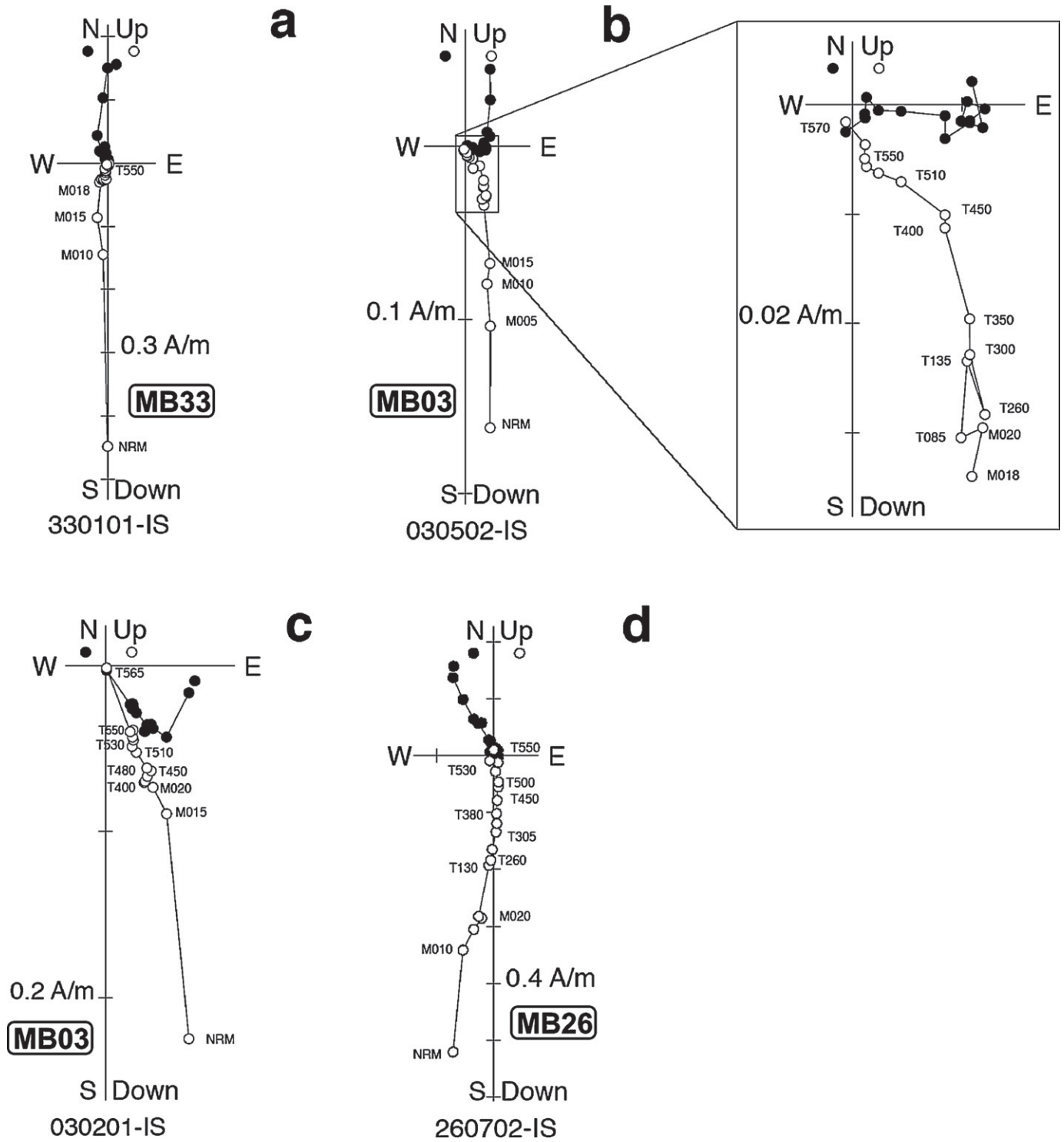


Fig. 4. Orthogonal demagnetization vector plots (Zijderveld, 1967) for representative specimens from Baie des Moutons. See text for discussion. Filled circles mark the stepwise demagnetization vectors projected into the horizontal north–south, west–east plane (west–east, north–south plane for e–h). Open circles mark the same measured demagnetization steps projected into the vertical plane. Step numbers preceded by M refer to AF demagnetization steps in mT; T denotes thermal steps in degrees C; L refers to low temperature demagnetization steps (Dunlop and Argyle, 1991) in four liquid nitrogen cooling cycles; NRM is the natural remanent magnetization. Intensity of magnetization is given in A/m for each specimen as marked along scale ticks.

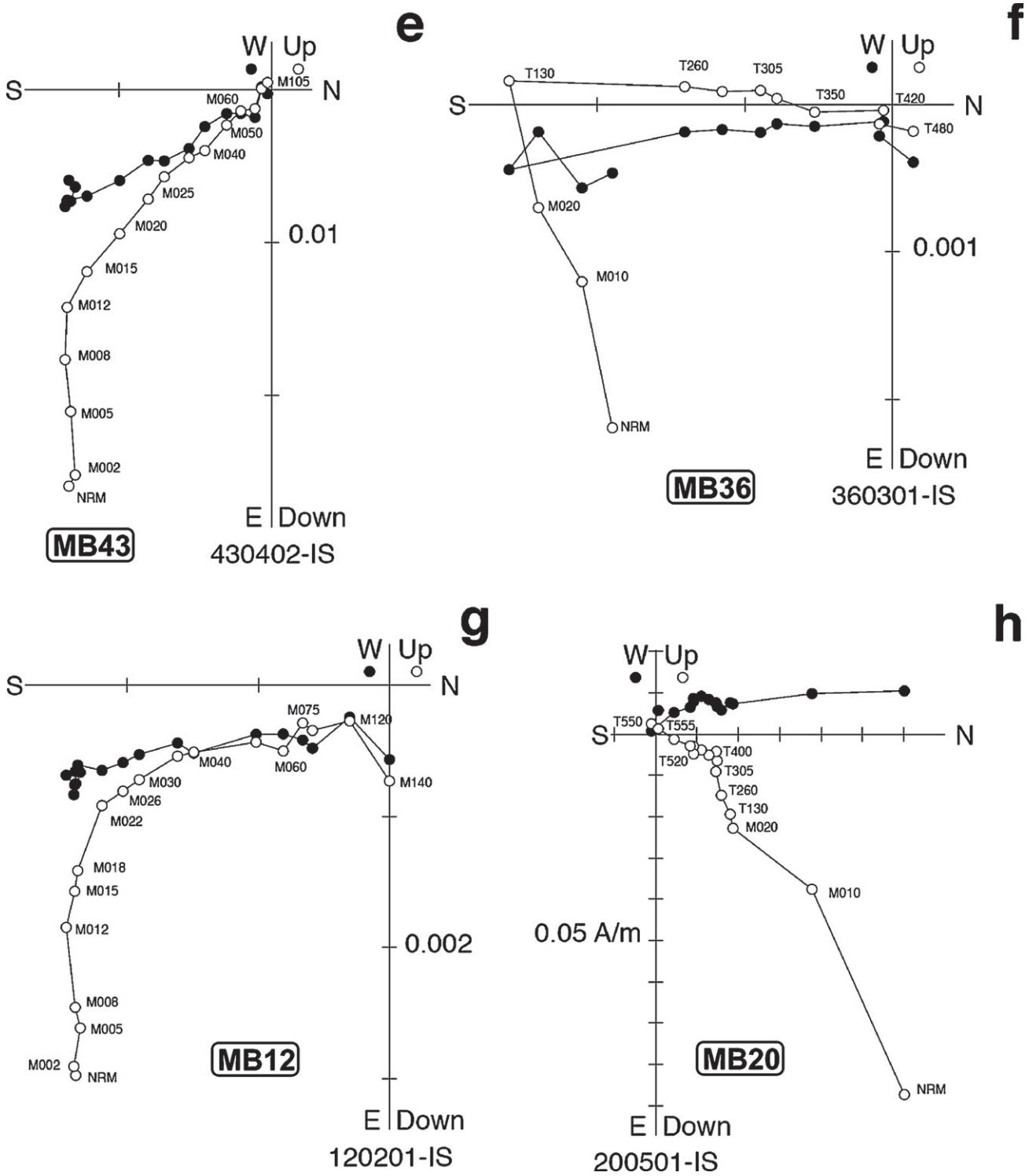


Fig. 4. (*Continued*)

however, that a direct comparison of a sister specimen from the same sample showed in many cases that the LTD-treated specimen carried a ChRM defined by high coercivity and high unblocking temperature over the final ~20% of the NRM, whereas the sister AF-cleaned specimen natural remanence consisted almost entirely of VRM with little or no ChRM present (Fig. 4k, l).

Disturbingly, the LTD pre-treatment appears to have created a spurious stable remanence which typically has a direction similar to the present Earth's field direction VRM that had been removed by the LTD! Given our acceptance criteria for ChRM A given above (remanence removed by demagnetization steps >450 °C or >20 mT), it is possible that LTD pre-treated specimens with a significant MD magnetite contribution may exhibit ChRM directions that are biased towards the present Earth's field direction. We have therefore excluded the LTD-treated specimens from further directional analysis in sites where AF demagnetization of other specimens revealed a large VRM component. The possible LTD modification of MD magnetite to produce a more stable remanence carrier is a

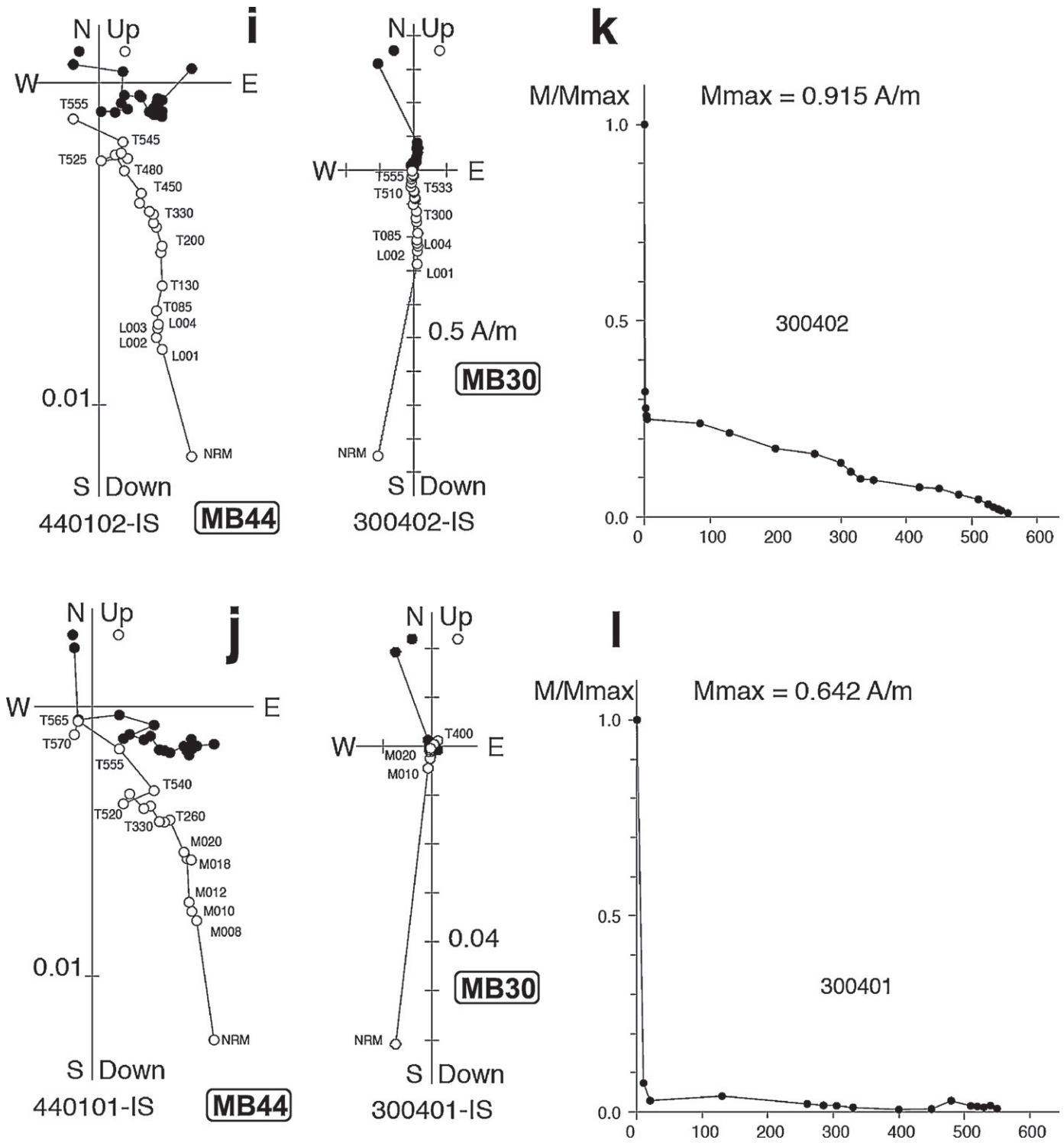


Fig. 4. (*Continued*).

phenomenon deserving of further study, but remains outside the scope of the present work.

3.4. ChRM A and ChRM B mean directions and tests for primary remanence

Site mean directions are defined on the basis of consistent ChRM directions from five or more specimens drawn from at least three samples at each site. We adopt an acceptance criterion of $\alpha_{95} < 15^\circ$ for the distribution of specimen ChRM directions.

Of the 44 sites sampled in the Baie des Moutons syenite, eight yielded consistent characteristic paleomagnetic ChRM A directions that were easterly and steep ($D = 98.6^\circ$ $I = 78.0^\circ$; $\alpha_{95} = 6.5^\circ$, $k = 71.7$), retained by PSD magnetite (Fig. 5b). Six sites from associated feldspar porphyry dykes and late intrusive units gave shallow southeasterly or northwesterly ChRM B directions ($D = 163.1^\circ$, $I = 6.0^\circ$; $\alpha_{95} = 21.7^\circ$, $k = 10.5$). These ChRM B sites can be broken down (Table 1) as four red feldspar porphyry dykes, or as five sites with the inclusion of a ribbon carbonatite dyke (site MB43). Note that for the six site case, the ChRM mean direction from late unit

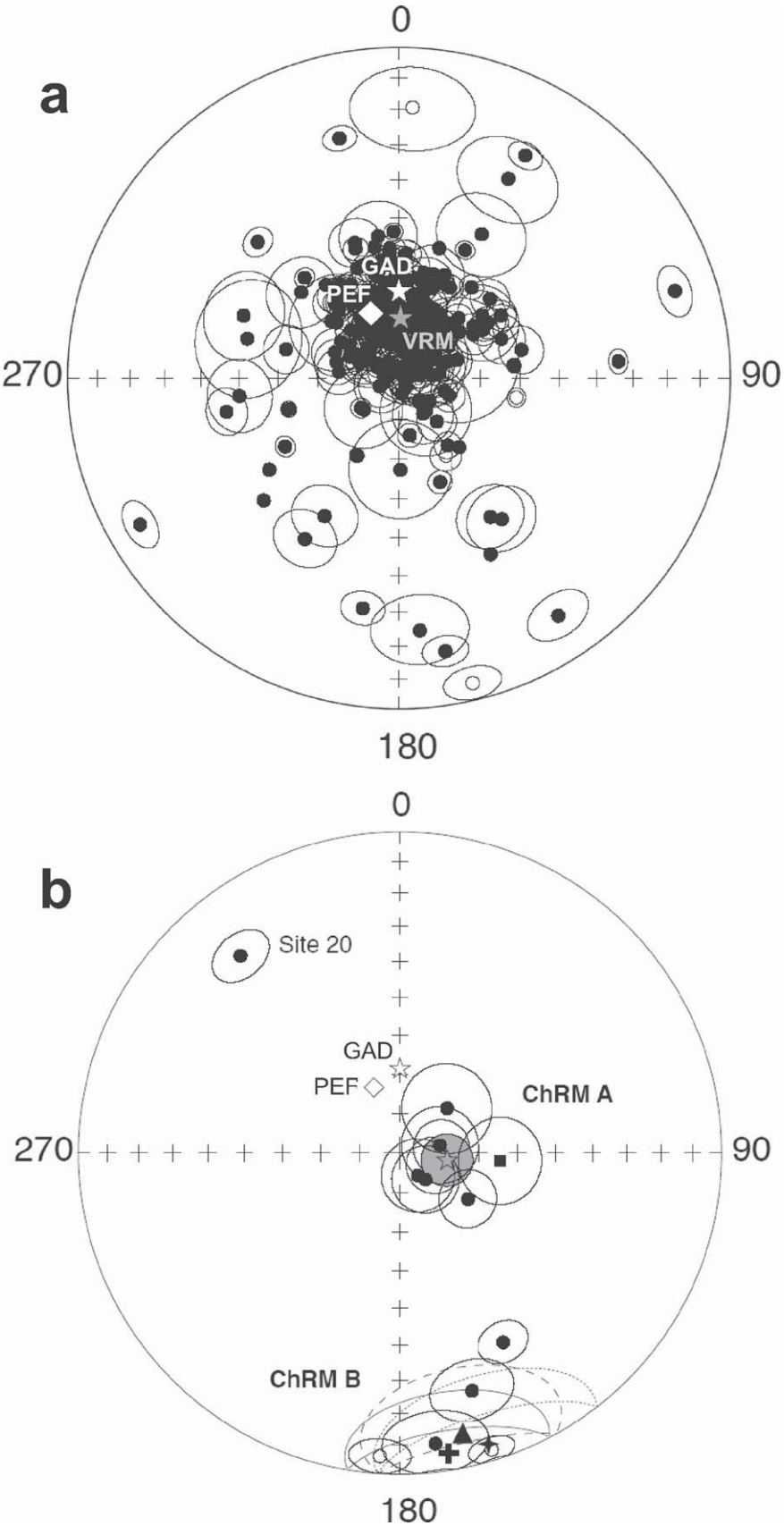


Fig. 5. Equal-area stereographic projections for a) VRM directions removed from 293 specimens, with the mean VRM shown as a grey star, and b) the site mean directions and their 95% confidence ellipses (Fisher, 1953; McFadden and McElhinny, 1988) for the Baie des Moutons syenite complex. The steep, easterly ChRM A mean direction is shown as a grey star with its 95% confidence ellipse; three shallow mean ChRM B means are shown with 95% confidence ellipses in broken lines (four point star is for $n = 4$ dykes sites only; triangle with dashed ellipse is for $n = 5$ sites, including the site MB43 carbonatite dyke; plus sign is for $n = 6$ sites, incorporating the inverted site MB20 mean direction). In both plots GAD (star) refers to the expected geocentric axial dipole field direction for the Baie des Moutons location and PEF (diamond) refers to the present-day Earth's field direction at Baie des Moutons during the time of its collection.

Table 2
Summary of ^{40}Ar–^{39}Ar results shown in Fig. 6 for the Baie des Moutons syenite, Quebec.

Site, unit	Mineral[a]	Apparent plateau age ± 1σ error
MB44, unit 1	hb	582.5 ± 1.6 Ma
MB44, unit 1	hb	584.1 ± 1.6 Ma
MB44, unit 1	bt	591.6 ± 1.7 Ma
MB44, unit 1	bt	586.2 ± 1.7 Ma
MB44, unit 1	K-fsp	(579.5) total gas
MB44, unit 1	K-fsp	(583.1) total gas
MB20, unit 3	hb	582.0 ± 1.8 Ma
MB20, unit 3	hb	585.0 ± 1.8 Ma
MB20, unit 3	bt	583.1 ± 1.6 Ma
MB20, unit 3	bt	584.4 ± 1.6 Ma
MB20, unit 3	K-fsp	(574.1) total gas
MB20, unit 3	K-fsp	(565.3) total gas
MB36, dyke	hb	(577.8) total gas
MB36, dyke	hb	583.5 ± 4.2 Ma
MB36, dyke	K-fsp	(527.0) total gas

[a] Mineral is: hb, hornblende; bt, biotite; K-fsp, potassium feldspar.

3 syenite site MB20 has been inverted for inclusion with the other five ChRM B site means, implying the occurrence of at least one reversal during the acquisition of ChRM B at Baie des Moutons.

Several contact tests were sampled on cross-cutting dykes and syenitic host rock at sites MB13–14, MB15, MB16, MB22, MB26 and MB36. In all cases the baked contact tests are unfortunately inconclusive, due to the lack of a stable, non-VRM dominated remanence in the host syenite at each site.

4. ^{40}Ar–^{39}Ar geochronology

4.1. Laboratory methods

Drill core and chip samples from three hornblende- and biotite-bearing sites (MB20, MB36 and MB44) were crushed to produce hornblende, biotite and potassium feldspar grains for ^{40}Ar–^{39}Ar geochronology. Hand-picked separates of hornblende, biotite and potassium feldspar grains were rinsed with deionized water and dried. Single, non-aggregate grains of 1–2 mm size were then selected under a binocular microscope for further analysis. Several grains from each site were packaged at the University of Michigan Radiogenic Isotope Geochemistry Laboratory in an aluminum foil capsule and irradiated for 300 h at the Phoenix-Ford Memorial Reactor at the University of Michigan. Neutron flux gradients were monitored using the MMhb-1 standard (520.4 Ma; Samson and Alexander, 1987).

During analysis, single grains were step-heated until fusion by an automated argon-ion laser which was defocused to ensure uniform heating. Individual steps consisted of 60 s heating followed by 2 min of gas purification using two 10 l/s SAES getters (ST101 alloy) and a liquid N_2 cold finger to remove the active gasses prior to mass spectrometry. Argon isotopic ratios were measured with a VG1200S mass spectrometer. Extraction line blanks were run after every six steps. Duplicate grains were analyzed to assess grain-to-grain variability and the reproducibility of the results. The ^{40}Ar–^{39}Ar plateau ages given below are reported as 1σ and based on the Ar measured during at least five consecutive heating steps that represent >50% of the total gas released, with step ages that pass a chi-squared test of the squared weighted deviations from the weighted average age (McDougall and Harrison, 1999). Plateau ages with 1σ errors are calculated as the inverse variance weighted mean of the ages from the steps which comprise the plateau. Error contributions from decay constant and standard uncertainties have not been included. All argon isotopic data are set out in an online appendix table and are summarized in Table 2.

4.2. ^{40}Ar–^{39}Ar results and analysis

A sample from site MB44 green syenite under the foundation of Mr. Q. Lessard's house in La Tabatiére (Fig. 2), representing the first intrusive unit of the Baie des Moutons syenite, provided individual hornblende grains C80a and C80b for analysis. Fig. 6a and b depicts the apparent ages for both grains as a function of the stepwise cumulative fraction of total released ^{39}Ar. The two site MB44 hornblendes give replicate plateau ages of 584.1 ± 1.6 Ma (Fig. 6a) and 582.5 ± 1.6 Ma (not shown) that agree within error and with the total gas ages for each. The Ar stepwise release shows a clear relationship with composition (Cl/K, Ca/K ratios); despite the presence of small variations in Cl/K and Ca/K, the ages are remarkably consistent and suggest that one phase dominates the degassed Ar isotopic composition. The weighted mean ^{40}Ar–^{39}Ar plateau age of the two replicate hornblende grains from Mr. Lessard's basement is 583.3 ± 1.9 Ma.

Biotite grains from the same site MB44 green syenite sample were also analyzed; grain C81a provided a plateau age of 591.6 ± 1.7 Ma (Fig. 6b) and grain C81b gave a similar plateau age of 586.2 ± 1.7 Ma (not shown). The mean ^{40}Ar–^{39}Ar plateau age of the two replicate biotite grains is 588.9 ± 3.9 Ma, slightly older than their hornblende couterparts. Site MB44 potassium feldspar grains C82a (Fig. 6c) and C82b provide similar total gas ages (579.5 ± 1.8 and 583.1 ± 1.8 Ma, respectively) to that of the hornblendes. Progressive step heating of the feldspar grains give apparent ages which have minima of ca. 575 Ma and then an asymptotic approach to the ca. 583 Ma ages of the hornblendes, indicating that the feldspars have a simple history, likely having not experienced a thermal event since the mid-Ediacaran.

Unit 1 green syenite at site MB44 bears hornblende, biotite and potassium feldspar with similar ^{40}Ar–^{39}Ar ages of between 590 and 580 Ma, despite the differing closure temperatures for these minerals (hornblende, ~500 °C; biotite ~350 °C; potassium feldspar, ~280–200 °C; McDougall and Harrison, 1999), implying that unit 1 of the Baie des Moutons intrusion cooled rapidly following its emplacement.

Site MB20 is sampled from a red syenite roadcut outcrop (Fig. 2) representing unit 3, the last volumetrically major intrusive phase of Baie des Moutons (Lalonde and Martin, 1983). Replicate site MB20 hornblende grains B66a (Fig. 6d) and B66b (not shown) provide ^{40}Ar–^{39}Ar plateau ages of 582.0 ± 1.8 and 585.0 ± 1.8 Ma, respectively. The mean of these two hornblende ages is 583.5 ± 2.8 Ma. Biotite grains B68a (Fig. 6e) and B68b (not shown) have ^{40}Ar–^{39}Ar plateau ages of 583.1 ± 1.6 and 584.4 ± 1.6 Ma, respectively, indistinguishable from one another and from the hornblende ages. Potassium feldspar grains B67a (Fig. 6f) and B67b show degassing steps with minima at apparent ages of ~540 Ma, followed by an approach to ~570–580 Ma apparent ages for further ^{39}Ar release fractions. Hornblende, biotite and potassium feldspar grains from site MB20 behave similarly to those from the site MB44 green syenite and provide similar interpreted ^{40}Ar–^{39}Ar ages (Table 2).

The first and the last major intrusive units of Baie des Moutons have overlapping ^{40}Ar–^{39}Ar hornblende plateau ages, indicating that the main intrusive events likely took place at or slightly before 583.4 ± 2.0 Ma, the weighted mean of the four hornblende ^{40}Ar–^{39}Ar plateau ages from site MB44 and site MB20. The similarity of the biotite (586.2 ± 4.0 Ma; $n = 4$) and potassium feldspar apparent ages with those of the hornblendes further suggests that the entire syenitic body of Baie des Moutons was emplaced over only a few million years and subsequently cooled rapidly.

A "late" red feldspar porphyry dyke was sampled as site MB36 (Fig. 2) and provided hornblende and potassium feldspar separates. Hornblende grains C83b and C83c (Fig. 6g, h) exhibit variable composition and apparent ages over the initial 30% of ^{39}Ar release,

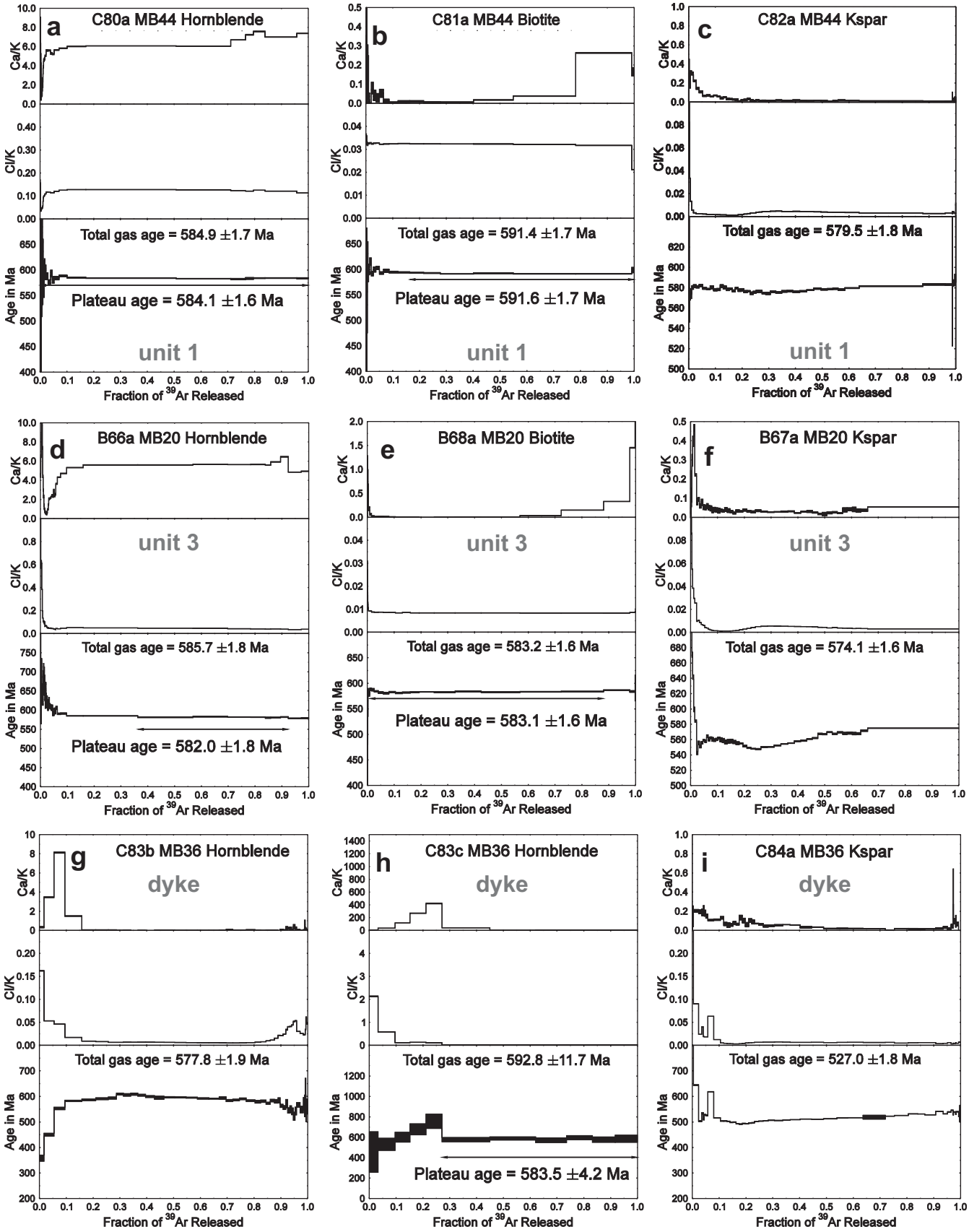


Fig. 6. Argon gas release spectra, plotted as apparent age versus cumulative ^{39}Ar gas release during laser step heating analysis at the University of Michigan. Contiguous degassing steps in hornblende or biotite that are spanned by an arrowed interval comprise the quoted plateau age at which closure of the K–Ar system is interpreted to have occurred; in all cases a total gas age is also calculated. Hornblende, biotite and potassium feldspar single crystal results are shown for grains from: a, b, c) Site MB44 (Unit 1) syenite; d, e, f) Site MB20 (Unit 3) syenite; and, g, h, i) Site MB36 red feldspar porphyry dyke.

but subsequently provide more stable degassing behaviour, giving an ^{40}Ar–^{39}Ar plateau age of 583.5 ± 4.2 Ma for the latter grain. A potassium feldspar grain from the dyke has a total gas age of 527.0 ± 1.8 Ma, with a minimum apparent age of ~500 Ma, followed by an approach to ~540 Ma (Fig. 6i), implying that the feldspar has experienced some loss of radiogenic ^{39}Ar during its history.

The site MB36 dyke is considered to be co-magmatic with, and a late product of the intrusion (Lalonde and Martin, 1983). Its hornblende ^{40}Ar–^{39}Ar ages are consistent with that interpretation, overlapping within error with the hornblende ages from Baie des Moutons intrusive units 1 and 3 (Table 2).

In summary, we interpret the magmatic hornblende, biotite and potassium feldspar grain analyses from all three sites to indicate that the Baie des Moutons syenite was emplaced at ca. 583 Ma, in mid-Ediacaran time, and subsequently underwent rapid cooling in a shallow rift setting.

5. Discussion

5.1. Acquisition ages of ChRM A and ChRM B

The intrusive age of Baie des Moutons may be best represented by the four hornblende ^{40}Ar–^{39}Ar plateau ages from the first and last major intrusive units, having a weighted mean of 583.4 ± 2.0 Ma. Biotite grains from the same intrusive units give ^{40}Ar–^{39}Ar plateau ages with a weighted mean of 586.2 ± 4.0 Ma (n = 4 grains), indistinguishable from those of the hornblendes. The step heating degassing behaviour of the hornblende and biotite grains is not complex (Fig. 6), suggesting that the plateau ages date the simple closure of the ^{40}K–^{39}Ar system at ~500 °C (hornblende) and ~350 °C (biotite; McDougall and Harrison, 1999). Potassium feldspars from both units also appear to be little disturbed by any subsequent heating or chemical alteration events. These ^{40}Ar–^{39}Ar data collectively imply that the Baie des Moutons syenite has not been significantly heated or subjected to chemical alteration since its mid-Ediacaran emplacement.

Characteristic remanent magnetizations (ChRMs) acquired during the initial cooling of the intrusion in the presence of the Earth's ca. 583 Ma paleofield would be expected to survive to the present-day as thermoremanent magnetizations, given magnetic mineralogy that records with sufficient fidelity (Pullaiah et al., 1975). In the present paleomagnetic study of the Baie des Moutons syenite, however, only 14 out of 44 sites were deemed to clearly have ChRMs carried by PSD magnetite, due mainly to the presence in most sites of MD magnetite that carries an intense VRM of recent origin (Figs. 4 and 5; Table 1).

Two distinct, apparently mutually exclusive directional ChRM groups were found in our study (Fig. 5b). Eight sites were found to carry a steep, ChRM A direction and six sites were found to have a shallow, south-southeasterly (or northwesterly) ChRM B direction. Unfortunately, six attempted baked contact tests on dykes carrying ChRM A or ChRM B directions yielded inconclusive results, leaving the age of remanence acquisition for both ChRM A and ChRM B an open question. We also cannot directly determine the relative acquisition timing of the two Baie des Moutons ChRM directions, given their mutually exclusive occurrence.

ChRM A and ChRM B may be distinguished, however, by the relative timing of their host intrusive units (Table 1). Seven of the eight ChRM A bearing sites come from Baie des Moutons intrusive units 1 or 2 (Fig. 2) and one from a mafic dyke cutting unit 3, whereas ChRM B is found elsewhere in the final portion of intrusive unit 3, an aplite dyke in unit 1 and in sites from north-northeast trending dykes that are the last known intrusive activity at Baie des Moutons (Davies, 1968; Lalonde, 1981; Lalonde and Martin, 1983). The simplest implied order of remanence acquisition is the steep ChRM A first, followed by shallow ChRM B acquisition during the waning magmatic stages of the intrusion. The overlapping ^{40}Ar–^{39}Ar ages which bracket the entire intrusive activity at Baie des Moutons would imply, however, that if ChRMs A and B are thermoremanent magnetizations, then they must essentially be coeval. Indeed, all three dated sites carry resolved ChRMs: site MB44 in unit 1 bears ChRM A; site MB20 in unit 3 bears a northwesterly ChRM B; and, the red feldspar porphyry NNE dyke at site MB36 carries ChRM B.

ChRM A is steeply directed to the east, but is still close to the PEF direction and exhibits only the downwards polarity (Fig. 5). It has, however, been determined using more stringent demagnetization methods designed to screen out PEF influence (e.g., Dunlop and Argyle, 1991), and so is unlikely to be a result of incomplete removal of a viscous component parallel to the PEF. ChRM A does not otherwise resemble any directions expected at the Baie des Moutons location from the Phanerozoic apparent polar wander path (APWP) for North America (Van der Voo, 1993); the steep ChRM A is not likely to be an overprint magnetization.

The shallow ChRM B is dissimilar to the more easterly shallow directions found in late Ediacaran–early Cambrian Laurentia (McCausland et al., 2007), but is similar to the shallow directions found in the 564 Ma Sept Iles intrusion (Sept Iles 'A' of Tanczyk et al., 1987; Raub et al., 2009), and also to Early Carboniferous results from Laurentia. As with the case for the much-disputed Sept Iles A magnetization (Tanczyk et al., 1987; Symons and Chiasson, 1991; Meert et al., 1994; Murphy et al., 2004; Cawood and Pisarevsky, 2006; McCausland et al., 2007; Raub et al., 2009), the shallow Baie des Moutons ChRM B exists in two polarities and may also be a thermoremanent magnetization.

Remanence acquisition age for ChRM A and ChRM B at Baie des Moutons is most simply interpreted in each case as being primary to the rapid cooling of the intrusion following its ca. 583 Ma emplacement. Any further distinction between the acquisition age of ChRMs A and B requires the successful execution of field tests for primary remanence or the identification of plausible remagnetization mechanisms.

5.2. Baie des Moutons paleopoles

The eight sites bearing ChRM A directions yield a paleopole "BMa" at 42.6°N, 332.7E (dp = 11.7°, dm = 12.4°) which places Laurentia at high paleolatitudes (Fig. 7; Table 3). The six sites with ChRM B directions give a paleopole "BMb" at −34.2°N, 321.5E (dp = 10.9°, dm = 21.8°), indicating that Laurentia lay at low paleolatitudes when ChRM B was acquired. Four of the six ChRM B sites are drawn from NNE-trending dykes (Table 1); these four sites alone provide a tighter paleopole at −35.5°N, 312.3E (dp = 8.5°, dm = 17.0°). The inclusion in the N = 6 mean of ChRM B means from site MB43 – a ribbon carbonatite dyke – and site MB20 – a "late" unit 3 syenite – is deemed appropriate given that these sites appear to have similar mean directions to the four dykes and also occur "late" in the Baie des Moutons magmatic sequence (Davies, 1968; Lalonde and Martin, 1983).

We assign a quality (Q) factor (Van der Voo, 1990, 1993) of 5 out of 7 for the BMa paleopole from the steep ChRM A; it fails criterion #4 (field test constrains age of remanence) and #6 (presence of reversals). The shallow ChRM B paleopole BMb is deemed to also have Q = 5; it has a reversed site, but fails criterion #4 and #7 (no resemblance to younger paleopoles).

5.3. Laurentia APW during the Ediacaran

Ediacaran through Cambrian paleopoles for Laurentia are given in Table 3, drawn mainly from the Global Paleomagnetic database v. 4.5 (Pisarevsky, 2005), noting minor updates. The entries are filtered to retain those results having a Q-factor (Van der Voo,

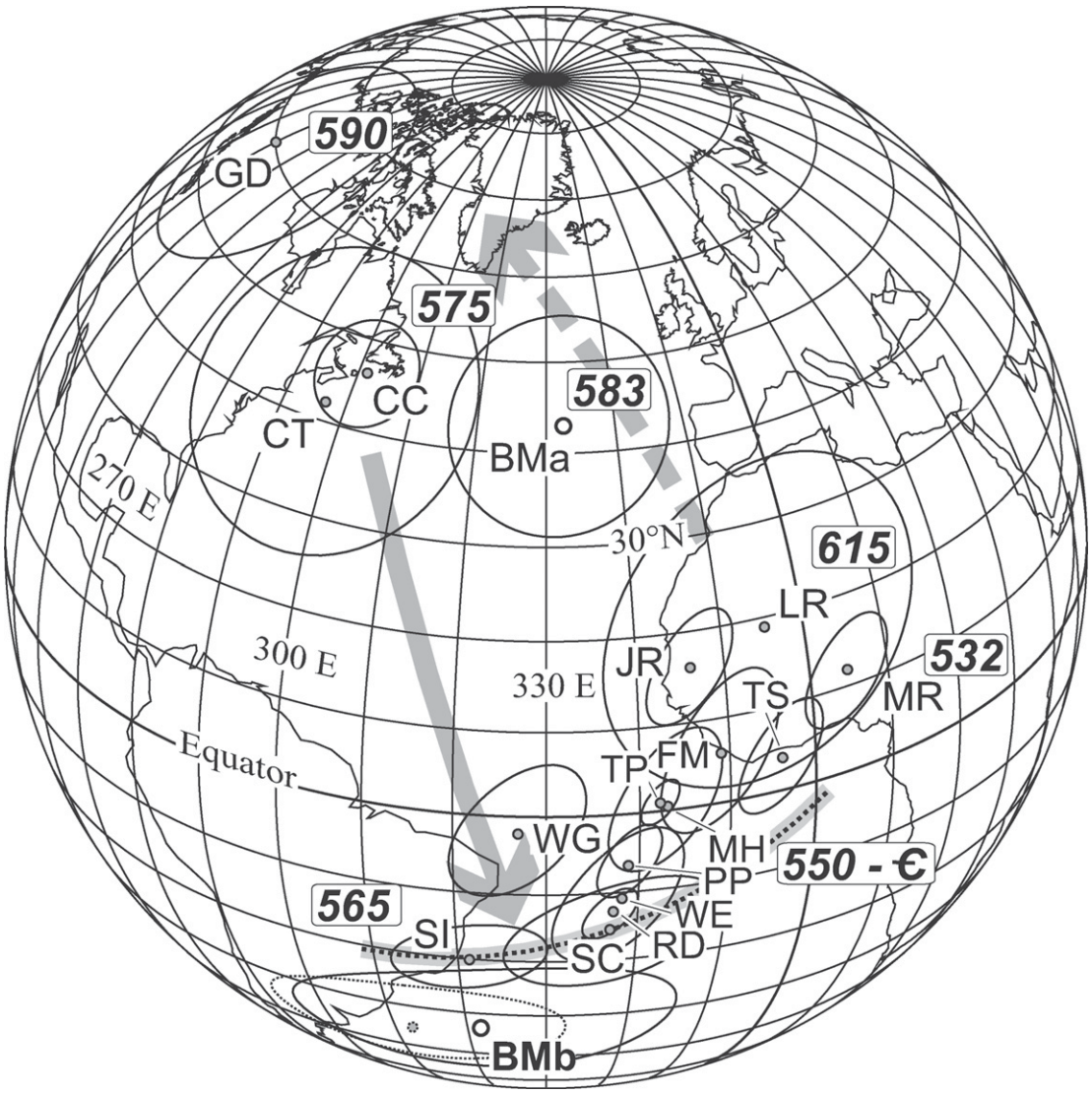


Fig. 7. Ediacaran to Cambrian paleopoles for Laurentia, plotted as south poles with respect to present-day North America. Unit abbreviations and references are given in Table 3. The curved, dashed swath through the Skinner Cove (SC) paleopole indicates the possible range of ca. 550 Ma SC pole positions, given that it comes from an allochthonous unit in Western Newfoundland that is of Laurentian origin (Cawood et al., 2001; Hodych et al., 2004), but may have experienced local net vertical axis rotation. The large grey arrows denote rapid APW of an Ediacaran loop for Laurentia; the dashed arrow shows Early to Mid-Ediacaran APW, from the 615 Long Range dykes paleopole to the 590 Ma Grenville dykes paleopole, whereas the solid arrow shows mid to late Ediacaran APW from the ca. 575 Ma poles to the ca. 564 Ma Sept Iles (SI) result, or the ca. 550 Ma Skinner Cove (SC) result.

1990, 1993) of 3 out of 7 or greater. A detailed discussion of these accepted Laurentian paleopoles is given in McCausland et al. (2007); only relevant updates will be discussed below. The Laurentian Ediacaran–Cambrian paleopoles are plotted in Fig. 7 as south poles, following the convention of McCausland et al. (2007).

Laurentia's Ediacaran through Cambrian APW continues to be the subject of controversy (Symons and Chiasson, 1991; Meert et al., 1994; McCausland and Hodych, 1998; Pisarevsky et al., 2001, 2008; Meert and Torsvik, 2003; Evans, 2003; Hodych et al., 2004; Murphy et al., 2004; Cawood and Pisarevsky, 2006; McCausland et al., 2007; Abrajevitch and Van der Voo, 2010) concerning the evidence for, timing of, and possible mechanisms responsible for rapid APW of Laurentia. In Fig. 7 and Table 3, the paleopoles for Laurentia fall into two groupings, one swathe of poles which place Laurentia at low paleolatitudes in 615 Ma and again at 565 Ma and through the Cambrian, and a second group of 590–570 Ma poles which place Laurentia at high paleolatitudes. McCausland et al. (2007) regarded these paleopoles as describing a large, distinctive APW feature – an Ediacaran loop – that implies that Laurentia moved to high paleolatitudes between 615 and 590 Ma and then back to low latitudes between 570 and 565 Ma.

The large APW for Laurentia is directly represented by large paleolatitude shifts which could be considered to be due to rapid Laurentia plate motion in excess of 30 cm/a (McCausland and Hodych, 1998), or large-scale true polar wander (TPW), involving the bulk tumbling of the Earth with respect to its spin axis (Deutsch, 1963; Golderich and Toomre, 1969; Evans, 2003), or some combination of both mechanisms. In principle, a substantial TPW contribution to the apparent motions of continents should appear uniquely in the paleomagnetic record as simultaneous, long APW tracks for all parts of the Earth (Deutsch, 1963). McCausland et al. (2007) compared the Ediacaran paleomagnetic record for Laurentia, Baltica, Siberia and elements of West Gondwana, concluding that TPW could not be ruled out, given the low number of available well-dated paleopoles and the large Ediacaran APW for each continent.

Abrajevitch and Van der Voo (2010) have proposed an alternative explanation for the two groupings of Laurentia Ediacaran

Table 3
Ediacaran to Cambrian paleomagnetic results for Laurentia.

	Rock unit, location	N	Pole (south)		A_{95} or	Q	Unit age (Ma)	REFNO
			N	E	*dp/dm*			
LR	Long Range dykes, Labrador	5	19.0	355.3	14.8/20.5	5	615 ± 2[a]	2657[*]
GD	Grenville Dykes 'B', QC	12	60.0	240.0	14.0	3	590 ± 2[b]	682
BMa	Baie des Moutons A	8	42.6	332.7	11.7/12.4	5	583 ± 2	This study
BMb	Baie des Moutons B	6	−34.2	321.5	10.9/21.8	5	583 ± 2	This study
CC	Callander Alkaline Complex, ON	26	46.3	301.4	5.9/6.1	6	577 ± 1[b]	2508
CT	Catoctin Basalts 'A', VA	6	42.0	296.7	17.4/17.7	6	572 ± 5[c]	2868
SI	Sept Iles intrusion 'A', QC	10	−20.1	321.3	4.9/8.8	4	564 ± 4[d]	1925
JR	Johnnie Fm., NV (unrotated)	~15	−10.0	342.0	5.5/11.0	4	ca. 555[e]	1216
SC	Skinner Cove Fm, NFLD	10	−16.4	337.5	7.0/11.0	4	550 ± 3[f]	3214
WG	Wichita granites, OK	18	−2.0	327.0	6.0/9.0	4	533 ± 3[g]	523
MR	Mont Rigaud & Chatham-Grenville	15	11.9	4.6	3.8/6.9	5	532 ± 2	*a*
TP	Tapeats sandstone, AZ (unrotated)	6	−4.9	337.7	1.7/3.3	4	513–501	1044
FM	Florida Mountains auriole, NM	22	6.0	349.0	8.0	6	503 ± 10	2375
TS	Taum Sauk limestone, MO	4	−4.0	356.0	3.6/7.2	5	501–488	1284
MH	Moore Hollow Gp, TX	10	0.6	343.0	4.5/8.7	4	501–488	2383
WE	Welge ss, Wilberns Fm, TX	9	−10.8	338.2	5.0/9.9	5	501–488	801
PP	Point Peak ls, Wilberns Fm, TX	7	−6.4	338.8	2.4/4.7	5	501–488	2376
RD	Royer Dolomite, OK	~6	−12.6	337.3	2.2/4.4	6	501–472	2289

N: Number of paleomagnetic sites in the given study which define its result, with approximate values (~) determined by dividing the number of reported contributing samples by six; *Q*: Quality value (maximum of 7; Van der Voo, 1990); Unit age is given as a radiometric age with ± error or as a stratographic age, with references numbered and given below for ages not appearing in the paleomagnetic reference; REFNO is the paleomagnetic result reference in the Global Paleomagnetic Database (version 4.6; Pisarevsky, 2005); result *a* (McCausland et al., 2007) is not in database.

[*] Long Range dykes as recalculated by Hodych et al. (2004).
[a] Age references Kamo et al. (1989), Kamo and Gower (1994).
[b] Kamo et al. (1995).
[c] Aleinikoff et al. (1995).
[d] Higgins and van Breemen (1998).
[e] Meert et al. (1994), but see Hodych et al. (2004) and McCausland et al. (2007).
[f] Cawood et al. (2001).
[g] Wright et al. (1996).

paleopoles, invoking the occurrence of an equatorial dipole geomagnetic field (Aubert and Wicht, 2004). They also note that similar steep and shallow directional groupings occur for Baltica during the Ediacaran, pointing out that with an equatorial dipole, the steep directions need not represent high paleolatitudes for either Baltica or Laurentia. In their equatorial diploe hypothesis, the 'shallow' and 'steep' directional groupings for Laurentia and Baltica may be accounted for by geologically instantaneous jumps from a normal geocentric axial dipole to an equatorial dipole geometry, respectively, and back again (Abrajevitch and Van der Voo, 2010). Unlike rapid plate motions (Meert et al., 1993) or TPW (Tsai and Stevenson, 2007), the equatorial dipole mechanism does not have a geological speed limit and so can readily explain rapid APW. The analysis by Abrajevitch and Van der Voo (2010) perhaps overstates the rapidity of APW for Laurentia by drawing upon marginal results giving shallow directions for the ca. 590 Ma Grenville Dykes (Murthy, 1971), steep directions for the ca. 564 Ma Sept Iles intrusion (Tanczyk et al., 1987; McCausland et al., 2007; Raub et al., 2009) and shallow directions for the Catoctin basalts (Cactoctin 'B') for which the remanence age is poorly constrained (Meert et al., 1994; McCausland et al., 2007), but the mechanism they propose is nevertheless applicable to – and testable with – well-constrained Ediacaran paleopoles.

In the present study, we have found two distinct directional groups in syenites and dykes from the same ca. 583 Ma Baie des Moutons intrusive event. As noted above, baked contact tests for ChRMs A and B proved inconclusive, so it is not possible to rule out either Baie des Moutons ChRM as being primary. It remains possible that both ChRM A and ChRM B are primary and of the same age; if this is true then rapid plate motion and TPW can be excluded as explanations for the large APW and appeal must be made to some form of non-GAD geometry for the ancient Earth geomagnetic field, such as the equatorial dipole proposed by Abrajevitch and Van der Voo (2010).

5.4. Assumptions for Ediacaran paleogeography

In the face of such disparate paleopole data and possible non-uniformitarian mechanisms, some prejudices must be maintained – at least until proven otherwise. Our preference in the discussion below is to choose polarity options for the paleopoles so as to place them in the same hemisphere for each source continent during the Ediacaran. West Gondwana is assumed to have been an essentially intact entity on one plate by ca. 630 Ma (Pimentel et al., 2000; Reno et al., 2009; Cordani et al., 2009), although some interpretations hold that the assembly of Amazonia with West Africa and Congo-Sao Francisco did not take place until Cambrian time (Trindade et al., 2006; Tohver et al., 2010). Finally, in the paleogeographic options offered below we strive to maintain the approximate Laurentia, Baltica and West Gondwana relationships to match those known from Cambro-Ordovician time (e.g., Meert and Lieberman, 2004). We also restrict our analysis to these presumed circum-Iapetus continents, ignoring in the present study the Ediacaran large APW record from Australia (Schmidt and Williams, 2010) and Siberia (Kravchinsky et al., 2001; Li, 2008) as well as isolated results from other continents and terranes.

In this analysis, we take the approach of attempting to identify coherent patterns in the APW records of Laurentia, Baltica and elements of West Gondwana, as simple APW path-fitting (Fig. 8) or with an imposed initial paleogeography of Iapetus opening (e.g, Bingen et al., 1998; Murphy et al., 2004) (Fig. 9). Our assessment of the APW records leads us to propose two paleogeographic options for the Ediacaran: (1) paleogeography that drops out of a path fit of 615–580–560 Ma paleopole groups (Fig. 8); (2) paleogeography that assumes a Laurentia–Baltica–West Gondwana triple junction opening of the Iapetus (Fig. 9a) with a high latitude Laurentia during the mid-Ediacaran (Fig. 9b) or a similar Iapetus opening paleogeography but with Laurentia residing at low latitudes throughout the Ediacaran (Fig. 9a, c).

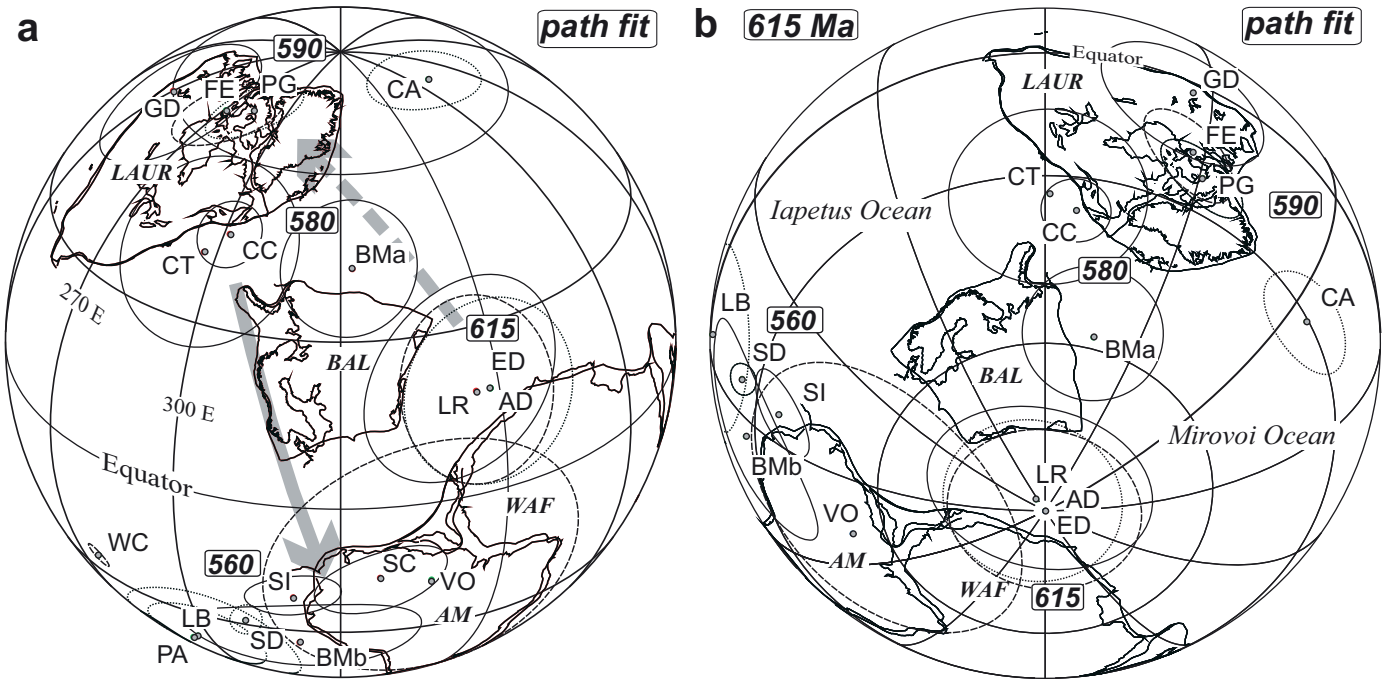


Fig. 8. Ediacaran APW loop path fit paleogeography in a) Laurentian co-ordinates with grey arrows as in Fig. 7 and, b) as a paleogeography for the Early Ediacaran (615 Ma). Paleopoles for Laurentia (solid lines), Baltica (dashed) and West Gondwana (dotted) are shown with 2σ uncertainties, plotted as groupings at 615, 590, 580 and 560 Ma, using unit abbreviations from Tables 3 and 4. Note that the remanence ages of some results are disputed (e.g., for the Puga carbonate; Trindade et al., 2003 and the Fen complex; Meert et al., 1998), as discussed in the text.

5.5. Ediacaran path fit paleogeography

Regarding Ediacaran paleogeography, the recognition of a pattern of large APW tracks for many continents may render the problem solvable if a sufficient number of well-defined paleopoles with good age constraints become available for comparison, and the assumption is made that the large APWs do not represent rapid plate motions for the continents, but rather the coherent signature of large-scale TPW (Evans, 2003; McCausland and Van der Voo, 2004) or of switching between a geocentric axial dipole (GAD) field and a geographically fixed equatorial dipole (Abrajevitch and Van der Voo, 2010).

Fig. 8 depicts a candidate Ediacaran paleogeography for the circum-Iapertus region based on the assumption that the Ediacaran paleopoles from Laurentia, Baltica and elements of West Gondwana are all primary, showing distinct APW tracks which take the form of an Ediacaran loop (McCausland et al., 2007). It is similar to the paleogeography shown by McCausland and Van der Voo (2004) based on TPW tracks and that of Abrajevitch and Van der Voo (2010) using paired "steep" and "shallow" directional groupings because these reconstructions are conceptually the same despite invoking different mechanisms for the APW. Notably, reconstructions based on APW track matching have the potential to constrain relative paleolongitudes (Van der Voo, 1993); this property is of particular usefulness for the Ediacaran, a period which likely features dispersed continents following the breakup of Rodinia and prior to the final assembly of Gondwana (Tohver et al., 2006; Li, 2008).

Three main groupings of poles are shown in Laurentian coordinates in Fig. 8a, one for ~615 Ma, one for 590–580 Ma and one for ~560 Ma (Tables 3 and 4). The ages of some paleopoles in each group are the subject of debate, but the general trend recognizable from Laurentia, Baltica and West Gondwana is that of a distinct Ediacaran loop (McCausland et al., 2007). Matching the 615–590–560 Ma loop for each continent results in a permissible paleogeography which has Baltica placed between Laurentia and the Amazonia-West Africa craton margin of West Gondwana, broadly similar to their Cambrian relations (Meert and Lieberman, 2004).

Notably, four paleopoles in the ~560 Ma grouping are actually interpreted to be of mid-Ediacaran age. These results are the Volhyn (580 Ma; Elming et al., 2007), Pan de Azucar (579 Ma; Sanchez-Bettucci and Rapalini, 2002; Oyhantçabal et al., 2007), Nola Metadiorite (571 Ma; Moloto-Kenguemba et al., 2009) and Baie des Moutons B (583 Ma; this study). Taken at face value, these paleopoles are difficult to reconcile with the other mid-Ediacaran paleopoles even using interpretations that invoke rapid plate motion or TPW (Abrajevitch and Van der Voo, 2010).

Fig. 8b shows the 'path fit' paleogeography for 615 Ma, with an interpreted south pole position based on paleomagnetic results from the 615 Ma Long Range dykes (Murthy et al., 1992; recalculated by Hodych et al., 2004) for Laurentia (Table 3), the 613 Ma Adma diorite (Morel, 1981) for West Gondwana, and the 616 Ma Egersund dykes (Walderhaug et al., 2007) for Baltica (Table 4). We caution, however, that use of the Adma diorite paleopole may be questionable because it has no test for primary remanence and a similarity to mid-Cambrian results for West Gondwana; the Adma diorite is also located just east of the West Africa craton, in the Pharusian fold belt (Morel, 1981), and so may not directly represent the craton or West Gondwana (Tohver et al., 2006).

The "path fit" paleogeography depicted in Fig. 8 is strikingly dissimilar to any shown previously for the latest Precambrian, in that it implies that Laurentia, Baltica and West Gondwana were well isolated from one another. In particular, the South American margin of West Gondwana is located far away from the Iapetan margin of Laurentia, implying that the two continents did not have the conjugate rift margin relationship during the breakup of Rodinia that is usually interpreted from geological relationships (Dalziel, 1997; Bingen et al., 1998; Murphy et al., 2004; Li, 2008). Depicted relations between Baltica and West Gondwana are arguably correct, placing the Colombia through Morocco portion of Amazonia and West Africa cratons opposite the southwestern and southeastern margins of Baltica (Murphy et al., 2004 and references therein).

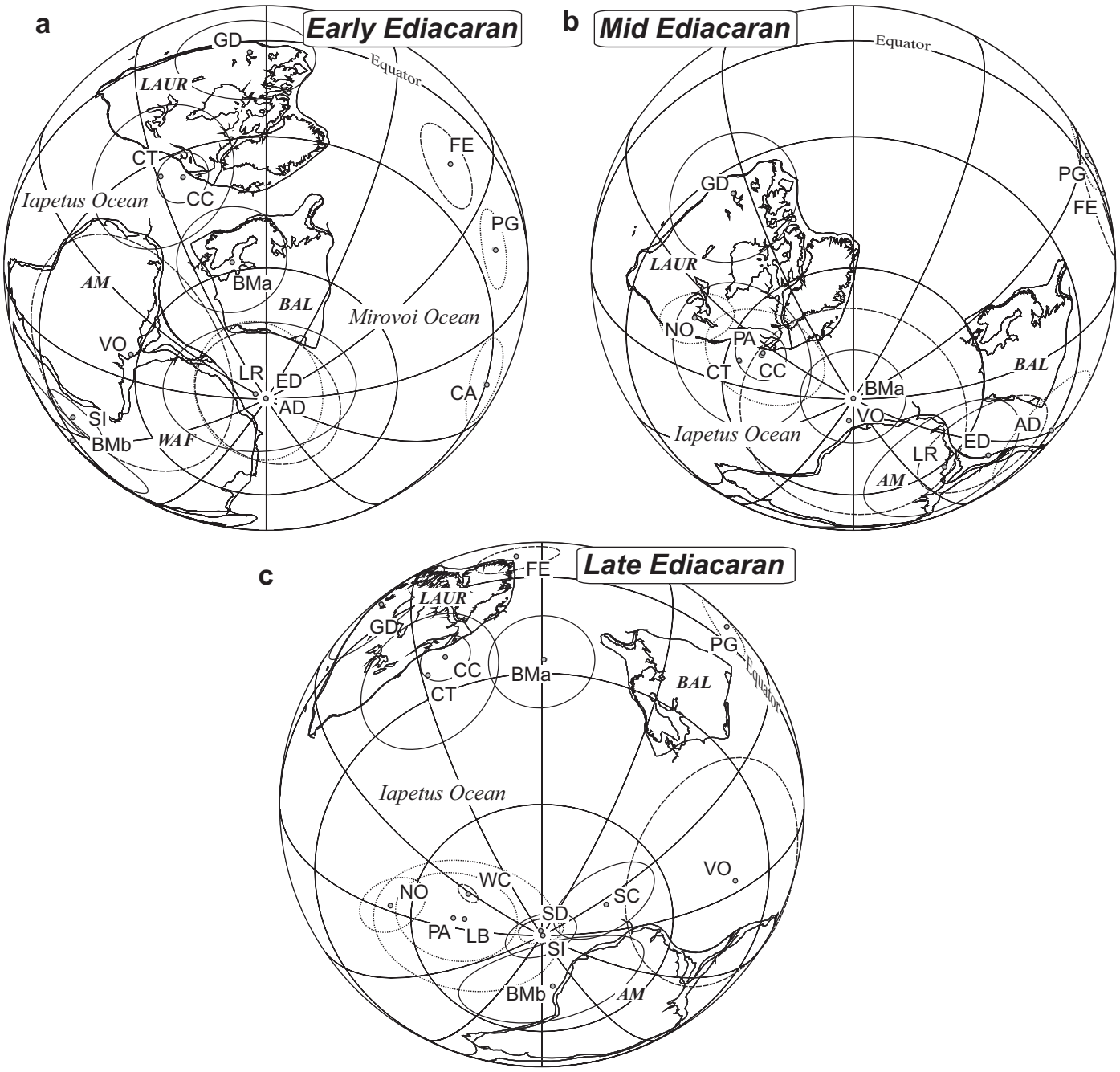


Fig. 9. Ediacaran paleogeography based on assuming an initial Laurentia–Baltica–West Gondwana triple rift geometry (e.g., Bingen et al., 1998; Murphy et al., 2004). Paleogeography for the circum-Iapetus region is shown for: a) the early Ediacaran (~615 Ma); b) the mid-Ediacaran (~580 Ma); and c) the late Ediacaran (post 565 Ma). The paleogeography in c) also can be taken to represent the "low latitude" Laurentia option (Pisarevsky et al., 2008), in which Laurentia may have resided at low latitudes throughout the Ediacaran. Paleopoles are plotted with 2σ uncertainties as solid lines for Laurentia, dashed lines for Baltica and dotted lines for West Gondwana, drawn from Tables 3 and 4. Euler pole rotations for the Laurentia, Baltica and West Gondwana reconstructions are given in Table 5.

Relations between Baltica and Laurentia are geologically awkward, leaving the incipient Iapetan rift margin of Laurentia facing the convergent Timanide margin of Baltica at 615 Ma (Cawood and Pisarevsky, 2006).

The path-fitting technique employed in reaching the paleogeography in Fig. 8 has been used in other circumstances to predict the "distal" paleolocation of continents (e.g. for Siberia with respect to Laurentia in the late Mesoproterozoic; Pisarevsky and Natapov, 2003). Here, we do not think that Laurentia's distal position with respect to Baltica and West Gondwana as shown in Fig. 8b is geologically viable. The paleogeography is a product of the assumptions that: (1) all of the paleopoles used are primary; and, (2) relative plate motions were minimal and that nearly all of the APW was due to some other mechanism such as TPW or an unusual geomagnetic field geometry. In seeking alternatives to the "path fit" endmember paleogeography of Fig. 8, we must then relax one or both of these assumptions.

5.6. Early Ediacaran Iapetan rift Ediacaran paleogeography

Another Ediacaran paleogeography may be found by adopting a new assumption, that the interpretation of post-Rodinia rift relations between Laurentia, Baltica and West Gondwana (Bingen et al., 1998; Murphy et al., 2004) is correct and offers a starting point for

Table 4
Ediacaran paleomagnetic results for West Gondwana and Baltica.

	Rock unit, location	*N*	Pole (south)		A_{95} or	*Q*	Unit age (Ma)	REFNO
			N	E	*dp/dm*			
	West Gondwana (reported in African coordinates*, from Tohver et al., 2006)							
AD	Adma diorite, Mali (WAF-Pharus)	13	33.5	343.5	16.0/17.0	*3*	613 ± 3	1170
PG	Puga cap carbonate (AM)	13	51.0	249.0	5.6/9.3	*4*	~600	3669
CA	Campo Alegre rhyolites (LA)	6	82.0	233.6	9.2/13.6	*6*	598 ± 29	1369
PA	Pan de Azucar syenite (RP)	6	−46.8	312.0	15.9/21.5	*4*	579 + 3/−1	3513
NO	Nola metadolerite (C-SF)	9	−61.8	304.8	5.4/10.7	*4*	571 ± 6	–a–
LB	Los Barrientos claystones (RP)	12	−44.2	312.7	10.9/14.2	*3*	~550	–b–
SD	Sinyai metadolerite, Kenya (Moz)	~8	−28.0	319.0	3.0/5.0	*5*	547 ± 4	3106
	Baltica (modified from Meert et al., 2007)							
ED	Egersund dykes, Norway	9	31.4	44.1	14.4/17.1	*5*	616 ± 3	447, 689
FE	Fen carbonatite complex, Norway	6	53.2	145.7	5.6	*4*	583 ± 15	3447
VO	Volhyn basalts, Ukraine	5	19.8	004.4	28.3	*4*	580 ± 4	–c–
WC	Winter Coast shale, Russia	15	31.0	293.0	9.9	*6*	555 ± 1	3506

Rock unit locations are given regional modifiers (*in italics*) for West Gondwana, recognizing that it may not have been fully assembled throughout Ediacaran–Cambrian time (Tohver et al., 2006). Abbreviations are: AM – Amazonia; Brazil – Braziliano belt; C-SF – Congo-Sao Franciso; LA – Luís Alves; Moz – Mozambique belt; RP – Rio de la Plata; WAF-Pharus – West Africa-Pharusian belt. *N*: Number of paleomagnetic sites in the given study which define its result, with approximate values (~) determined by dividing the number of reported contributing samples by six; *Q*: Quality value (maximum of 7; Van der Voo, 1990, 1993); Unit age is given as a radiometric age with ± error or as a stratographic age; REFNO is the paleomagnetic result reference in the Global Paleomagnetic Database (version 4.6; Pisarevsky, 2005). Letters in the REFNO column mark results which are not in the database, and are referenced as follows: a – Moloto-Kenguemba et al. (2009); b – Rapalini (2006); c – Elming et al. (2007), Nawrocki et al. (2004).

* South America rotated to Africa (WAF) coordinates by the Euler rotation pole 45.5°N, 327.8E, 58.2° CCW (Lawver and Scotese, 1987).

Ediacaran paleogeography, leading to the opening of the Iapetus Ocean between those continents. Fig. 9a shows a similar geologically viable starting point for 615 Ma, based on the paleopoles from the Long Range dykes, Adma diorite and Egersund dykes (Tables 3 and 4).

The essential difference between the "path fit" paleogeography of Fig. 8b and the "Iapetan rift" paleogeography of Fig. 9a is the location of Laurentia, which in the latter figure is shown shifted in paleolongitude so as to be in rift relationship with Baltica and Amazonia. The relative position of Laurentia and Baltica is similar to that inferred for the early Neoproterozoic Valhalla orogen (Cawood et al., 2009) between the two cratons and places the 615 Ma Long Range dykes of Laurentia in regional proximity to the coeval Egersund dykes of Baltica (Walderhaug et al., 2007). The rift relationship between Laurentia and Amazonia is assumed, there being no direct evidence for Ediacaran rifting along the west margin of Amazonia (Cordani et al., 2003, 2009); rifting along Amazonia was likely mediated instead through terranes outboard of Amazonia, with the leading candidates being Arequipa-Antofallia (Cordani et al., 2009) or Oaxaquia (Murphy et al., 2004; Keppie et al., 2006).

5.7. Mid-Ediacaran high latitude Laurentia

A consequence of the revised location of Laurentia is that several mid-Ediacaran paleopoles from Baltica and West Gondwana no longer match those from Laurentia. This is evident in Fig. 9b for ~580 Ma, which has a paleogeography driven mainly by assuming moderate 615–580 Ma relative plate motions between Laurentia, Baltica and West Gondwana. Mid-Ediacaran Laurentia paleopoles that imply a high paleolatitude (such as the BMa result from this study) are used to fix the south pole. We interpret two mid-Ediacaran paleopoles from West Gondwana, Pan de Azucar in Rio de la Plata craton (Sanchez-Bettucci and Rapalini, 2002; 579 Ma age from Oyhantçabal et al., 2007) and the 571 Ma Nola Metadiorite in Congo craton (Moloto-Kenguemba et al., 2009) to indicate that the proto-Andean margin of South America lay in the polar region at 580–570 Ma, approximately opposite Laurentia across a widening Iapetus Ocean (Fig. 9b). For Baltica, the 580 Ma Volhyn basalts paleopole (Nawrocki et al., 2004; Elming et al., 2007) suggests that Baltica also lay near the polar region in a 'rightside-up' orientation (Cawood and Pisarevsky, 2006).

This mid-Ediacaran configuration of Iapetus-bounding continents leads us to reject results from the 583 Ma Fen complex in Baltica (Meert et al., 1998) and ~600 Ma results from West Gondwana (Puga and Campo Alegre; Table 4) because all three poles plot together near the paleoequator, almost 90° from the depicted 580 Ma pole position (Fig. 9b). Puga (Trindade et al., 2003) and Fen (Meert et al., 1998) poles have difficulty demonstrating primary remanence and could be a record of later overprints (Kilner et al., 2005; Pisarevsky et al., 2008). The Campo Alegre (D'Agrella-Filho and Pacca, 1988) paleopole is well-determined but may not be tectonically coherent with West Gondwana at ~600 Ma (Trindade et al., 2006). Results from the Huqf supergroup (Kilner et al., 2005) in the Arabian–Nubian Shield are not used here

Table 5
Euler pole rotations for continents in Figs. 8 and 9.

Euler operation	Euler pole, rotation		
	N	E	CCW
Fig. 8: Ediacaran APW path fit			
WGond to Laurentia:	14.71	212.28	−59.79
Baltica to Laurentia:	12.09	133.70	24.42
Laurentia to absolute reference frame, using the Long Range dykes pole (LR in Table 3):	2.05	265.30	−109.00
Fig. 9a: 615 Ma			
WGond to Laurentia:	19.19	342.95	−86.40
Baltica to Laurentia:	66.44	247.45	−43.80
Laurentia to absolute reference frame, using the Long Range dykes pole (LR in Table 3):	2.05	265.30	−109.00
Fig. 9b: 583 Ma			
WGond to Laurentia:	4.76	353.67	−109.75
Baltica to Laurentia:	49.16	005.01	−71.87
Laurentia to absolute reference frame, using the Baie des Moutons pole (BMa in Table 3):	0	242.70	−132.60
Fig. 9c: 560 Ma			
WGond to Laurentia:	24.36	145.31	89.00
Baltica to Laurentia:	9.88	061.18	45.98
Laurentia to absolute reference frame, using the Sept Iles pole (SI in Table 3):	0	051.30	69.90

Note: WGond refers to West Gondwana cratons in African co-ordinates; see Table 4 for details.

because they suffer from poor control of remanence age as well as an unknown Ediacaran coherence with West Gondwana. Alternatively, in the 580 Ma circum-Iapetus paleogeography of Fig. 9b, the three 'rejected' paleopoles might represent the activity of an equatorial dipole (Abrajevitch and Van der Voo, 2010), although there is no corresponding paleopole from Laurentia to join their grouping.

5.8. Low latitude Laurentia

Fig. 9c shows late Ediacaran paleogeography for 565–545 Ma, based on the 565 Ma Sept Iles (Tanczyk et al., 1987; Raub et al., 2009) and 550 Ma Skinner Cove (McCausland and Hodych, 1998; Hodych et al., 2004) results for Laurentia (Table 3), the ca. 550 Ma Los Barrientos siltstones (Rapalini, 2006) and the 547 Ma Sinyai metadolerite dykes (Meert and Van der Voo, 1996) for West Gondwana (Table 4) and the Winter Coast result for Baltica (Popov et al., 2002, 2005).

If the Baie des Moutons B (BMb) paleopole is taken to be primary, Laurentia would have resided at low paleolatitudes in the mid-Ediacaran as well as in the late Ediacaran, with similar pole positions given by the 583 Ma Baie des Moutons, 565 Sept Iles and 550 Ma Skinner Cove results. This is the "low latitude" Laurentia alternative, where Laurentia is interpreted to have been at low paleolatitudes throughout the Ediacaran (Murphy et al., 2004; Hodych et al., 2004; Pisarevsky et al., 2008). Note that the two West Gondwana ~550 paleopoles are also similar to the 579 Ma Pan de Azucar and 571 Ma Nola Metadiorite paleopoles; in Fig. 9c they collectively place the Amazonia portion of West Gondwana near the pole and well separated across a wide Iapetus Ocean from the low latitude Laurentia throughout the mid and late Ediacaran.

In the low latitude Laurentia interpretation (Murphy et al., 2004; Hodych et al., 2004; Pisarevsky et al., 2008) the mid-Ediacaran Laurentian paleopoles from the Callander complex (Symons and Chiasson, 1991), Catoctin basalts (Meert et al., 1994), Grenville dykes (Murthy, 1971), and from Baltica, the Fen complex (Meert et al., 1998) are either Paleozoic overprints or mixtures of an ancient remanence with some contribution from the steep present Earth's field in the sampling regions in North America and northern Europe (Hodych et al., 2004).

A study of the 584 ± 7 Ma Alnø carbonatite complex of Sweden (Meert et al., 2007) revealed a high unblocking temperature, high coercivity remanence which had scattered low-inclination directions broadly similar to the low-inclination result reported by Piper (1981), but also a grouping of site mean directions indistinguishable from that of the present-day field. Meert et al. (2007) noted that their definition of low- and high-inclination site mean directions was rather arbitrary, being drawn from a streaked distribution of high to low-inclination site means. They considered both the low- and high-inclination results to be of uncertain utility in Ediacaran paleogeographic analyses (Meert et al., 2007) and so we have excluded them from consideration here.

In our present study of the Baie des Moutons syenite, we find a major contribution from a VRM that records the steep northerly present Earth field direction, but also find a steeply directed high unblocking temperature and high coercivity remanence which provides the paleopole BMa (Table 3; Figs. 8 and 9). If the steep directions in Baie des Moutons, Alnø, Fen, Callander, Grenville dykes and Catoctin are interpreted as carrying a partial contribution from the present Earth's field, then a magnetization mechanism must be identified to explain this occurrence.

5.9. Tectonic and geodynamic implications

Given the wide variety of Ediacaran paleomagnetic results from Laurentia, Baltica and West Gondwana, it is difficult to make definitive conclusions regarding the paleogeography or geodynamics of that period. We have outlined options that highlight coherent APW tracks, with or without the assumption of an initial circum-Iapetus paleogeography. None of the paleogeographic options are entirely uniformitarian!

The "path fit" paleogeography requires the action of large-scale TPW or a non-GAD field geometry during the Ediacaran, with only minor contributions from relative plate motions. Both the "path fit" (Fig. 8) and the "high latitude" Laurentia (Fig. 9a, b) paleogeographic options appear to require coherent plate motions or perhaps TPW of at least 30° over 35 myr from 615 to ~580 Ma. The implied motion is ~10 cm/a, which is geodynamically reasonable for either plate motions (Meert et al., 1993) or for TPW (Tsai and Stevenson, 2007). Subsequent APW is large, however, for the "path fit" case, amounting to over 60° of motion over 20 myr (yielding a rate of nearly 40 cm/a) from ~580 to 560 Ma. For the "high latitude" Laurentia case (Fig. 9), post-580 Ma apparent motion is similarly high (>40 cm/a) and must be ascribed mostly to plate motion, given that Baltica and West Gondwana do not share the large APW during that 580–560 Ma period if the Puga, Campo Alegre and Fen poles are disregarded (Fig. 9b).

The possible occurrence of a plume under the Iapetan margin of Laurentia (Kumarapeli, 1993; Higgins and van Breemen, 1998; Puffer, 2002), has geodynamic significance in that the 590–550 Ma duration of plume-related activity implies that Laurentia remained nearly stationary with respect to the mantle during most of the Ediacaran. Located with a fixed plume, Laurentia could not have undergone large plate motions with respect to the mantle reference frame during the mid to late Ediacaran (Pisarevsky et al., 2008), but large TPW motion (Fig. 8) remains possible because the plume would move with the continent.

For the "low latitude" Laurentia paleogeography, relative plate motions are minor for Laurentia, Baltica and West Gondwana during the mid to late Ediacaran (Fig. 9c), but larger APW is implied for Laurentia, Baltica and West Gondwana during the early Ediacaran. For instance, 61° of APW is required from the 615 Ma Long Range dykes paleopole to the 583 Ma "low latitude" Baie des Moutons B paleopole. A similar APW for West Gondwana is found from comparison of the 613 Ma Adma diorite paleopole to the 579 Ma Pan de Azucar paleopole (Table 4), although it is possible that the host regions for West Gondwana had not yet assembled by 615–580 Ma (Trindade et al., 2006). As interpreted in Fig. 9a and c, the large APW is dominantly accounted for by plate motions, with West Gondwana moving over 60° (~6600 km) from Laurentia during the 615–580 Ma interval, marking a rapid, early Ediacaran opening of the Iapetus Ocean. Baltica has lesser but still significant relative plate motion with respect to Laurentia and West Gondwana.

Considering the assembly of West Gondwana, several authors have interpreted the paleogeography to maintain a Laurentia–Amazonia link until the late Ediacaran (Trindade et al., 2006; Tohver et al., 2010), showing a wide ocean (the Clymene Ocean; Trindade et al., 2006) between the Amazonia craton and the Rio de la Plata, Congo-Sao Francisco and West African cratons. We note that this option is equally viable to the ones presented here because there are currently no relevant Ediacaran paleomagnetic data to constrain the position of Amazonia. Assuming a Laurentia–Amazonia link until the late Ediacaran still does have severe geodynamic consequences, however, in that much of the wide gap between Laurentia and West Gondwana at 550 Ma (Fig. 9c) must be traversed by Amazonia during the closure of a wide Clymene Ocean between 550 and 525 Ma (Trindade et al., 2006; Tohver et al., 2010), in conjunction with an equally rapid opening of the Iapetus Ocean between Amazonia and Laurentia during the same period.

5.10. Will the real Ediacaran paleopole please stand?

From the foregoing analysis it is clear that the circumstances of the opening of the Iapetus Ocean and assembly of Gondwana remain poorly constrained. Ediacaran paleogeography is not solved with the results from the 583 Ma Baie des Moutons syenite or in our wider discussion of the paleomagnetic data from Laurentia, Baltica and elements of West Gondwana. The only clear conclusion regarding the Iapetus Ocean paleogeography to be found in these data is that it must have opened well before to 550 Ma (Cawood et al., 2001; McCausland et al., 2007), if the widely used triple junction Laurentia–Baltica–Amazonia initial paleogeography is assumed (Bingen et al., 1998; Murphy et al., 2004).

Assessment of the existing Ediacaran paleopole data appears to leave little room for a uniformitarian Earth during that period. The Ediacaran paleopole database may well be drastically revised in the coming years, perhaps resulting in less extreme APW. In all the paleogeographies discussed here with reference to the present database, some appeal to large TPW, rapid plate motions or a non-GAD geomagnetic field must be made. As befits an apparently extreme period in Earth's biotic and climate history (Grotzinger et al., 1995; Hoffman and Schrag, 2002; Knoll et al., 2004; Budd, 2008), the late Neoproterozoic also appears to present extreme geodynamic alternatives. The Ediacaran may be even stranger – and of more fundamental interest – than has previously been supposed.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.precamres.2011.02.004.

References

Abrajevitch, A., Van der Voo, R., 2010. Incompatible Ediacaran paleomagnetic directions suggest an equatorial geomagnetic dipole hypothesis. Earth and Planetary Science Letters 293, 164–170.

Aleinikoff, J.N., et al., 1995. U-Pb ages of metarhyolites of the Catoctin and Mount Rogers Formations, central and southern Appalachians: evidence for two pulses of Iapetan rifting. American Journal of Science 295, 428–454.

Aubert, J., Wicht, J., 2004. Axial vs equatorial dipolar dynamo models with implications for planetary magnetic fields. Earth and Planetary Science Letters 221, 409–419.

Bingen, B., Demaiffe, D., van Breemen, O., 1998. The 616 Ma old Egersund basaltic dike swarm, S.W. Norway and Late Neoproterozoic opening of the Iapetus Ocean. Journal of Geology 106, 565–574.

Budd, G.E., 2008. The earliest fossil record of the animals and its significance. Philosophical Transactions of Royal Society B 363, 1425–1434.

Cawood, P.A., McCausland, P.J.A., Dunning, G.R., 2001. Opening Iapetus: constraints from the Laurentian margin in Newfoundland. Geological Society of America Bulletin 113 (4), 443–453.

Cawood, P.A., Pisarevsky, S.A., 2006. Was Baltica right-way-up or upside-down in the Neoproterozoic? Journal of the Geological Society London 163, 1–7.

Cawood, P.A., van Gool, J.A.M., Dunning, G.R., 1996. Geological development of eastern Humber and western Dunnage zones: Corner Brook – Glover Island region Newfoundland. Canadian Journal of Earth Sciences 33, 182–198.

Cawood, P.A., Strachan, R., Cutts, K., Kinny, P.D., Hand, M., Pisarevsky, S.A., 2009. Neoproterozoic orogeny along the margin of Rodinia: Valhalla orogen North Atlantic. Geology 38, 99–102.

Cogné, J.P., 2003. PaleoMac: A MacintoshTM application for treating paleomagnetic data and making plate reconstructions. Geochemical and Geophysical Geosystem 4 (1), 1007, doi:10.1029/2001GC000227.

Cordani, U.G., D'Agrella-Filho, M.S., Brito-Neves, B.B., Trindade, R.I.F., 2003. Tearing up Rodinia: the Neoproterozoic palaeogeography of South American cratonic fragments. Terra Nova 15, 350–359.

Cordani, U.G., Teixeira, W., D'Agrella-Filho, M.S., Trindade, R.I.F., 2009. The position of the Amazonian Craton in supercontinents. Gondwana Research 15, 396–407.

Cox, R.A., Hodych, J.P., 2005. LA-ICP-MS zircon dating of the Lac Matapedia and Mt. St.-Anselme basalts, Quebec: support for a protracted mantle plume event during the opening of Iapetus. In: Geological Association of Canada – Mineralogical Association of Canada Joint Meeting , Halifax 30, p. 38.

D'Agrella-Filho, M.S., Pacca, I.G., 1988. Paleomagnetism of the Itajaí Castro and Bom Jardim Groups from southern Brazil. Geophysical Journal 93, 365–376.

Dalziel, I.W.D., 1997. Neoproterozoic-Paleozoic geography and tectonics: review, hypothesis, environmental speculation. Geological Society of America Bulletin 190 (1), 16–42.

Davies, R., 1968. Geology of the Mutton Bay intrusion and surrounding areas, North shore, Gulf of St. Lawrence, Quebec. Unpublished Ph.D. thesis, McGill University.

Deutsch, E.R., 1963. Polar wandering and continental drift: an evaluation of recent evidence. In: Munyan, A.C. (Ed.), Polar Wandering Continental Drift. Society of Economic Paleontologists Mineralogists, Tulsa, pp. 4–46.

Doig, R., Barton, J.M., 1968. Ages of carbonatites and other igneous rocks in Quebec. Canadian Journal of Earth Sciences 5, 1401–1407.

Dunlop, D.J., 2002. Theory and application of the Day plot (M_{rs}/M_s versus H_{cr}/H_c) 1. Theoretical curves and test using titanomagnetite data. Journal of Geophysical Research 107, doi:10.1029/2001JB000486.

Dunlop, D.J., Argyle, K.S., 1991. Separating multidomain and single-domain-like remanences in pseudo-single-domain magnetites (215-540 nm) by low-temperature demagnetization. Journal of Geophysical Research 96, 2007–2017.

Dunlop, D.J., Özdemir, O., 1997. Rock Magnetism: Fundamentals and Frontiers. Cambridge University Press, Cambridge, UK, 573 pp.

Elming, S.Å., Kravchenko, S.N., Layer, P., Rusakov, O.M., Glevasskaya, A.M., Mikhailova, N.P., Bachtadse, V., 2007. Paleomagnetism and 40Ar/39Ar age determinations of the Ediacaran traps from the southwestern margin of the East European Craton Ukraine: relevance to Rodinia break-up. Journal of Geological Society of London 164, 969–982.

Evans, D.A.D., 1998. True polar wander, a supercontinental legacy. Earth and Planetary Science Letters 157, 1–8.

Evans, D.A.D., 2003. True polar wander and supercontinents. Tectonophysics 362, 303–320.

Fisher, R.A., 1953. Dispersion on a sphere. Proceedings of the Royal Society of London A217, 295–305.

Gerencher, J.J., 1967. Structural relationships, petrography, chemistry and magnetic properties of some dike swarms in and around the Mutton Bay pluton, Quebec. Unpublished M.Sc. thesis, Pennsylvania State University.

Graham, J.W., 1949. The stability and significance of magnetism in sedimentary rocks. Journal of Geophysical Research 54, 131–167.

Golderich, P., Toomre, A., 1969. Some remarks on true polar wandering. Journal of Geophysical Research 74, 2555–2567.

Grotzinger, J.P., Bowring, S.A., Saylor, B.Z., Kaufman, A.J., 1995. Biostratigraphic and geochronologic constraints on early animal evolution. Science 270, 598–604.

Higgins, M.D., van Breemen, O., 1998. The age of the Sept Iles layered mafic intrusion, Canada: implications for the Late Neoproterozoic/Cambrian history of southeastern Canada. Journal of Geology 106, 421–431.

Hodych, J.P., Cox, R.A., Kosler, J., 2004. An equatorial Laurentia at 550 Ma confirmed by Grenvillian inherited zircons dated by LAM ICP-MS in the Skinner Cove volcanics of western Newfoundland: implications for inertial interchange true polar wander. Precambrian Research 129, 93–113.

Hoffman, P.F., 1991. Did the breakout of Laurentia turn Gondwanaland inside out? Science 252, 1409–1412.

Hoffman, P.F., Schrag, D.P., 2002. The snowball Earth hypothesis: testing the limits of global change. Terra Nova 14, 129–155.

Kamo, S.L., Heaman, L.M., Lumbers, S.B., 1989. Birthdate for the Iapetus Ocean? A precise U-Pb zircon and baddeleyite age for the Long Range dykes, southeast Labrador. Geology 17, 602–605.
Kamo, S.L., Gower, C.F., 1994. Note: U-Pb baddeleyite dating clarifies age of characteristic paleomagnetic remanence of Long Range dykes, southeastern Labrador. Atlantic Geology 30, 259–262.
Kamo, S.L., Krogh, T.E., Kumarapeli, P.S., 1995. Age of the Grenville dyke swarm Ontario-Quebec: implications for the timing of Iapetus rifting. Canadian Journal of Earth Sciences 32, 273–280.
Keppie, J.D., Dostal, J., Nance, R.D., Miller, B.V., Ortega-Rivera, A., Lee, J.K.W., 2006. Circa 546 Ma plume-related dykes in the ~1 Ga Novillo Gneiss (east-central Mexico): evidence for the initial separation of Avalonia. Precambrian Research 147, 342–353.
Kilner, B., MacNiocaill, C., Brasier, M., 2005. Low-latitude glaciation in the Neoproterozoic of Oman. Geology 33 (5), 413–416.
Kirschvink, J.L., 1980. The least squares line and plane and the analysis of paleomagnetic data. Geophysical Journal of the Royal Astronomical Society 62, 699–718.
Knoll, A.H., Walter, M.R., Narbonne, G.M., Christie-Blick, N., 2004. A new period for the Geologic Time Scale. Science 305, 621–622.
Kravchinsky, V.A., Konstantinov, K.M., Cogne, J.-P., 2001. Palaeomagnetic study of Vendian and Early Cambrian rocks of South Siberia and Central Mongolia: was the Siberian platform assembled at this time? Precambrian Research 110, 61–92.
Kumarapeli, P.S., 1993. A plume-generated segment of the rifted margin of Laurentia as seen through a completed Wilson Cycle. Tectonophysics 219, 47–55.
Kumarapeli, P.S., Dunning, G.R., Pintson, H., Shaver, J., 1989. Geochemistry and U-Pb zircon age of comenditic metafelsites of the Tibbit Hill Formation Quebec Appalachians. Canadian Journal of Earth Sciences 26, 1374–1383.
Kumarapeli, P.S., Saull, V.A., 1966. The St. Lawrence valley system: a North American equivalent of the East African rift valley system. Canadian Journal of Earth Sciences 3, 639–658.
Lalonde, A.E., 1981. The Baie-des-Moutons syenitic complex, La Tabatière, Quebec. Unpublished M.Sc. thesis, McGill University.
Lalonde, A.E., Martin, R.F., 1983. The Baie-des-Moutons syenitic complex, La Tabatière Quebec I. Petrography and feldspar mineralogy. Canadian Mineralogist 21, 65–79.
Lawver, L., Scotese, C.R., 1987. A revised reconstruction of Gondwanaland. In: McKenzie, G.D. (Ed.), Gondwana Six: Structure, Tectonics and Geophysics. American Geophysical Union Monograph, pp. 17–23.
Li, Z.X., et al., 2008. Assembly, configuration, and break-up history of Rodinia: a synthesis. Precambrian Research 160, 179–210.
Lukert, M.T., Banks, P.O., 1984. Geology and age of the Robertson River pluton. In: Bartholomew, M.J. (Ed.), The Grenville Event in the Appalachians and Related Topics. Geological Society of America, pp. 161–166.
McCausland, P.J.A., Hodych, J.P., 1998. Paleomagnetism of the 550 Ma Skinner Cove volcanics of western Newfoundland and the opening of the Iapetus Ocean. Earth and Planetary Science Letters 163, 15–29.
McCausland, P.J.A, Van der Voo, R., 2004. Rapid, coherent apparent polar wander in the terminal Neoproterozoic – Early Cambrian? EOS: Transactions of the American Geophysical Union 85 (47), Abstract U33A-0026.
McCausland, P.J.A., Van der Voo, R., Hall, C.M., 2007. Circum-Iapetus paleogeography of the Precambrian-Cambrian transition with a new paleomagnetic constraint from Laurentia. Precambrian Research 156, 125–152.
McDougall, I., Harrison, T.M., 1999. Geochronology and Thermochronology by the $^{40}Ar/^{39}Ar$ Method. Oxford University Press, Oxford, 269 pp.
McFadden, P.L., McElhinny, M.W., 1988. The combined analysis of remagnetization circles and direct observations in paleomagnetism. Earth and Planetary Science Letters 87, 161–172.
Meert, J.G., Lieberman, B.S., 2004. A paleomagnetic and paleobiogeographical perspective on latest Neoproterozoic and early Cambrian tectonic events. Journal of the Geological Society London 161, 477–487.
Meert, J.G., Torsvik, T.H., 2003. The making and unmaking of a supercontinent: Rodinia revisited. Tectonophysics 375, 261–288.
Meert, J.G., Van der Voo, R., 1996. Paleomagnetic and $^{40}Ar/^{39}Ar$ study of the Sinyai dolerite, Kenya: implications for Gondwana assembly. Journal of Geology 104, 131–142.
Meert, J.G., Van der Voo, R., Payne, T.W., 1994. Paleomagnetism of the catoctin volcanic province: a new Vendian Cambrian apparent polar wander path for North America. Journal of Geophysical Research 99, 4625–4641.
Meert, J.G., Torsvik, T.H., Eide, E.A., Dahlgren, S., 1998. Tectonic significance of the Fen Province, S. Norway: constraints from geochronology and paleomagnetism. Journal of Geology 106, 553–564.
Meert, J.G., Walderhaug, H.J., Torsvik, T.H., Hendriks, B.W.H., 2007. Age and paleomagnetic signature of the Alno carbonatite complex (NE Sweden): additional controversy for the Neoproterozoic paleoposition of Baltica. Precambrian Research 154, 159–174.
Meert, J.G., et al., 1993. A plate tectonic speed limit? Nature 363, 217–218.
Miller, B.V., Barr, S.M., 2004. Metamorphosed gabbroic dikes related to the opening of Iapetus Ocean at the St. Lawrence promontory: Blair River inlier, Nova Scotia, Canada. Journal of Geology 112, 277–288.
Moloto-Kenguemba, G.R., Trindade, R.I.F., Monié, P., Nédélec, A., Siqueira, R., 2009. A late Neoproterozoic paleomagnetic pole for the Congo craton: tectonic setting, paleomagnetism and geochonology of the Nola dike swarm (Central African Republic). Precambrian Research 164, 214–226.
Morel, P., 1981. Palaeomagnetism of a Pan-African diorite: a late Precambrian pole for western Africa. Geophysical Journal of Royal Astronomical Society 65, 493–503.
Murphy, J.B., Pisarevsky, S.A., Nance, R.D., Keppie, J.D., 2004. Neoproterozoic-Early Paleozoic evolution of peri-Gondwanan terranes: Implications for Laurentia-Gondwana connections. International Journal of Earth Sciences (Geol. Rundsch.) 93, 659–682.
Murthy, G.S., 1971. The paleomagnetism of diabase dykes from the Grenville Province. Canadian Journal of Earth Sciences 8, 802–812.
Murthy, G.S., Gower, C.F., Tubrett, M., Patzold, R., 1992. Paleomagnetism of Eocambrian Long Range dykes and Double Mer Formation from Labrador Canada. Canadian Journal of Earth Sciences 29, 1224–1234.
Nawrocki, J., Boguckij, A., Katanas, V., 2004. New late Vendian palaeogeography of Baltica and TESZ. Geological Quarterly 48, 309–316.
Oyhantçabal, P., Siegesmund, S., Wemmer, K., Frei, R., Layer, P., 2007. Post-collisional transition from calc-alkaline to alkaline magmatism during transcurrent deformation in the southernmost Dom Feliciano Belt (Braziliano-Pan-African Uruguay). Lithos 98, 141–157.
Pimentel, M.M., Fuck, R.A., Jost, H., Ferreira-Filho, C.F., Araujo, S.M., 2000. The basement of the Brasilia fold belt and the Goias magmatic arc. In: Cordani, U.G., Milani, E.J., Thomaz Filho, A., Campos, D.A. (Eds.), Tectonic Evolution of South America, 31st International Geological Congress Proceedings. , pp. 195–229.
Piper, J.D.A., 1981. Magnetic properties of the Alnø complex. Geologiska Foreningens Stockholm Forhandlingar 103, 9–15, Part 1.
Pisarevsky, S.A., 2005. New edition of the global paleomagnetic database. EOS Transactions 86, 170.
Pisarevsky, S.A., Komissarova, R.A., Khramov, A.N., 2001. Reply to comment by J.G Meert and R. Van der Voo on 'New paleomagnetic result from Vendian red sediments in Cisbaikalia and the problem of the relationship of Siberia and Laurentia in the Vedian. Geophysical Journal of International 146, 871–873.
Pisarevsky, S.A., Natapov, L.M., 2003. Siberia and Rodinia. Tectonophysics 375, 221–245.
Pisarevsky, S.A., Murphy, J.B., Cawood, P.A., Collins, A.S., 2008. Late Neoproterozoic and early Cambrian paleogeography: models and problems. In: Pankhurst, R.J., Truow, R.A., Brito Neves, B.B., de Wit, M.J. (Eds.), West Gondwana: Pre-Cenozoic Correlations Across the South Atlantic Region, 294. Geological Soc. Lond. Special Pub., pp. 9–31.
Popov, V., Iosifidi, A., Khramov, A.N., Tait, J., Bachtadse, V., 2002. Paleomagnetism of Upper Vendian sediments from the Winter Coast, White Sea region, Russia: implications for the paleogeography of Baltica during Neoproterozoic times. Journal of Geophysical Research 107 (B11), 2315, doi:10.1029/2001JB001607.
Popov, V., Khramov, A., Bachtadse, V., 2005. Palaeomagnetism, magnetic stratigraphy and petromagnetism of the Upper Vendian sedimentary rocks in the sections of the Zolotitsa River and in the Verkhotina Hole, Winter Coast of the White Sea, Russia. Russian Journal of Earth Science 7, 1–29.
Puffer, J.H., 2002. A late Neoproterozoic eastern Laurentian superplume: location, size, chemical composition, and environmental impact. American Journal of Science 302, 1–27.
Pullaiah, G.E., Irving, E., Buchan, K.L., Dunlop, D.J., 1975. Magnetization changes caused by burial and uplift. Earth and Planetary Science Letters 28, 133–143.
Rapalini, A.E., 2006. New late Proterozoic paleomagnetic pole for the Rio de la Plata craton: implications for Gondwana. Precambrian Research 147, 223–233.
Raub, T.D., Kirschvink, J.L., Higgins, M.D., Evans, D.A.D., Peek, S., Raub, T.M., 2009. Paleomagnetic restudy of the late Ediacaran Sept Iles mafic suite. EOS transactions, AGU. 90 (22) Toronto Joint Assembly Suppl., Abstract GA13B-05.
Reno, B.L., Brown, M., Kobayashi, K., Nakamura, E., Piccoli, P., AndTruow, R.A.J., 2009. Eclogite-high-pressure granulite metamorphism records early collision in West Gondwana: new data from the Southern Brasilia Belt Brazil. Journal of Geological Society 166, 1013–1032.
Samson, S.D., Alexander, E.C., 1987. Calibration of the interlaboratory 40Ar-39Ar dating standard MMhb-1. Chemical Geology 66, 27–34.
Sanchez-Bettucci, L., Rapalini, A.E., 2002. Paleomagnetism of the Sierra de las Animas complex, southern Uruguay: its implications in the assembly of Western Gondwana. Precambrian Research 118, 243–265.
Schmidt, P.W., Williams, G.E., 2010. Ediacaran palaeomagnetism and apparent polar wander path for Australia: no large true polar wander. Geophysical Journal of International, doi:10.1111/j.1365-246X.2010.04652.
Stampfli, G.M., Borel, G.D., 2002. A plate tectonic model for the Paleozoic and Mesozoic constrained by dynamic plate boundaries and restored synthetic oceanic isochrones. Earth and Planetary Science Letters 196, 17–33.
Steiger, R.H., Jaeger, E., 1977. Subcommission on geochronology: convention on the use of decay constants in geo- and cosmochronology. Earth and Planetary Science Letters 36, 259–262.
Stukas, V., Reynolds, P.H., 1974. 40Ar/39Ar dating of the Long Range dikes Newfoundland. Earth and Planetary Science Letters 22, 256–266.
Su, Q., Goldberg, S.A., Fullagar, P.D., 1994. Precise U-Pb zircon ages of Neoproterozoic plutons in the southern Appalachian Blue Ridge and their implications for the initial rifting of Laurentia. Precambrian Research 68, 81–95.
Symons, D.T.A., Chiasson, A.D., 1991. Paleomagnetism of the Callander complex and the Cambrian apparent polar wander path for North America. Canadian Journal of Earth Sciences 28, 355–363.
Tanczyk, E.I., Lapointe, P., Morris, W.A., Schmidt, P.W., 1987. A paleomagnetic study of the layered mafic intrusion at Sept-Iles Quebec. Canadian Journal of Earth Sciences 24, 1431–1438.
Tohver, E., D'Agrella-Filho, M.S., Trindade, R.I.F., 2006. Paleomagnetic record of Africa and South America for the 1200-500 Ma interval, and evaluation of Rodinia and Gondwana assemblies. Precambrian Research 147, 193–222.

Tohver, E., Trindade, R.I.F., Solum, J.G., Hall, C.M., Riccomini, C., Nogueira, A.C., 2010. Closing the Clymene ocean and bending a Brasiliano belt: evidence for the Cambrian formation of Gondwana, southeast Amazon craton. Geology 38, 267–270.

Tollo, R.P., Aleinikoff, J.N., Bartholomew, M.J., Rankin, D.W., 2004. Neoproterozoic A-type granitoids of the central and southern Appalachians: intraplate magmatism associated with episodic rifting of the Rodinian supercontinent. Precambrian Research 128, 3–38.

Trindade, R.I.F., D'Agrella-Filho, M.S., Epof, I., Brito-Neves, B.B., 2006. Paleomagnetism of Early Cambrian Itabaiana mafic dikes (NE Brazil) and the final assembly of Gondwana. Earth and Planetary Science Letters 244, 361–367.

Trindade, R.I.F., Font, E., D'Agrella-Filho, M.S., Nogueira, A.C.R., Riccomini, C., 2003. Low-latitude and multiple geomagnetic reversals in the Neoproterozoic Puga cap carbonate Amazon craton. Terra Nova 15, 441–446.

Tsai, V.C., Stevenson, D.J., 2007. Theoretical constraints on True Polar Wander. Journal of Geophysical Research 112 (B5), B05415, doi:10.1029/2005JB003923.

van Berkel, J.T., Currie, K.L., 1988. Geology of the Puddle Pond (12A/5) and Little Grand Lake (12A/12) map areas, southwestern Newfoundland. Newfoundland Department of Mines, Mineral Development Division report 88–1, pp. 99–107.

Van der Voo, R., 1990. The reliability of paleomagnetic data. Tectonophysics 184, 1–9.

Van der Voo, R., 1993. Paleomagnetism of the Atlantic Tethys and Iapetus Oceans. Cambridge University Press, Cambridge, 411 pp.

Waggoner, B., 1999. Biogeographic analyses of the Ediacara biota: a conflict with paleotectonic reconstructions. Paleobiology 25 (4), 440–458.

Walderhaug, H.J., Torsvik, T.H., Halvorsen, E., 2007. The Egersund dykes (SW Norway): a robust Early Ediacaran (Vendian) palaeomagnetic pole from Baltica. Geophysical Journal of International 168, 935–948.

Walsh, G.J., Aleinikoff, J.N., 1999. U-Pb zircon age of metafelsite from the Pinney Hollow Formation: implications for the development of the Vermont Appalachians. American Journal of Science 299, 157–170.

Williams, H., Gillespie, R.T, van Breemen, O., 1985. A late Precambrian rift-related igneous suite in western Newfoundland. Canadian Journal of Earth Sciences 22, 1727–1735.

Wright, J.E., Hogan, J.P., Gilbert, M.C., 1996. The southern Oklahoma aulacog: not just another B.L.I.P. EOS: Transactions of the American Geophysical Union 77 (46), F845.

Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: analysis of results. In: Collinson, D.W., Creer, K.M., Runcorn, S.K. (Eds.), Methods in Paleomagnetism. Elsevier, Amsterdam, The Netherlands, pp. 254–286.